Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

PHYSICS

Class – 10

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Table of Contents

S.No	Chapter Name	Page. No.
1	Light-Reflection	1 - 9
2	Light- Refraction	10 - 19
3	Human Eye and the Colourful World	20 - 26
4	CURRENT ELECTRICITY	27 - 38
5	MAGNECTIC EFFECTS OF ELCETRIC CURRENT	39 - 47

1. Light-Reflection

Solutions

Worksheet – 1

1. **Correct answer: D) Light cannot travel through a vacuum.**
   Light can travel through a vacuum; in fact, it does not need any material medium.
   Statements A, B, and C are correct descriptions of light.
   Hence, option D is the only wrong statement.
2. **Correct answer: B) Marginal rays**
   Rays that strike a curved mirror far from the principal axis and make large angles with it are called marginal rays.
   Paraxial rays, on the other hand, make small angles with the principal axis.
3. **Correct answer: D) In front of the mirror**
   A real image is formed when reflected rays actually meet.
   Such images can be obtained on a screen.
   In mirrors, this always happens in front of the mirror
4. **Correct answer: B) Virtual image**
   A virtual image is formed behind the mirror.
   The reflected rays only appear to meet and do not actually converge.
5. **Correct answer: B) Virtual and erect**
   A plane mirror always forms a virtual image that is upright (erect).
   The image is laterally inverted and of the same size as the object.
6. **Correct answer: C) Pole**
   The pole is the geometric centre of the reflecting surface of a spherical mirror.
   All distances related to the mirror are measured from the pole.
7. **Correct answer: B) Centre of curvature**
   The centre of curvature is the centre of the imaginary sphere of which the mirror forms a part.
   For a concave mirror, it lies in front; for a convex mirror, behind the mirror.
8. **Correct answer: B) The incident ray, reflected ray and the normal lie in the same plane**
   This statement represents the first law of reflection.
   It describes the geometric relationship between the incident ray, reflected ray, and normal.
9. **Correct answer: B) 35°**
   According to the second law of reflection, the angle of reflection equals the angle of incidence.
   Given angle of incidence = 35°.
   Therefore, the angle of reflection is also 35°.
10. **Correct answer: C) Paraxial rays**
    Paraxial rays are those rays that make small angles with the principal axis.
    They are close to the principal axis and are used in paraxial approximation.

Worksheet-2

1. **Correct Answer: (C) the principal focus**
   If a ray passes through the principal focus of a concave mirror, after reflection it travels parallel to the principal axis.
   This is one of the basic ray rules for spherical mirrors.
   Hence, the incident ray must have passed through the focus.
2. **Correct Answer: (B) At F**
   When an object is at infinity, the rays coming from it are parallel.
   In a concave mirror, parallel rays converge at the principal focus after reflection.
   So the image is formed at F.
3. **Correct Answer: (C) Between F and C**
   For an object placed beyond the centre of curvature (C),
   the image is formed between the focus (F) and centre of curvature (C).
   The image is real, inverted and smaller in size.

4. **Correct Answer: (B) Real and inverted**
When the object lies between F and C in a concave mirror,
the reflected rays meet in front of the mirror.
Thus, the image is real and inverted
5. **Correct Answer: (B) At infinity, real, inverted and highly magnified**
When the object is at the focus of a concave mirror,
the reflected rays become parallel to each other.
Hence, the image is formed at infinity and is highly magnified.
6. **Correct Answer: (B) real, inverted and diminished at focus**
For an object at infinity, parallel rays fall on the mirror.
These rays meet at the principal focus after reflection.
So the image is real, inverted and very small at F.
7. **Correct Answer: (C) Concave mirror**
ENT specialists use concave mirrors because they converge light.
This helps to focus more light on the ear, nose or throat.
Thus, the organs can be examined clearly.
8. **Correct Answer: (C) concave mirror**
A reflecting telescope uses a concave mirror.
It collects and focuses light from distant stars.
This produces a bright and clear image.
9. **Correct Answer: (C) the pole of the spherical mirror**
According to the new Cartesian sign convention,
all distances are measured from the pole (P) of the mirror.
This helps in applying mirror formulas correctly.
10. **Correct Answer: (B) positive**
In the new sign convention, distances measured upward
from the principal axis are taken as positive.
Downward distances are taken as negative.

Worksheet-3

1. **Correct answer is: C) R = 2f.**
For a spherical mirror under paraxial approximation,
the focal length is half the radius of curvature.
So, $f = \frac{R}{2}$ or $R = 2f$.
2. **Correct answer is: B) $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$**
The mirror formula for spherical mirrors is
$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$
3. **Correct answer is: C) for both convex and concave mirrors**
Validity of $f = \frac{R}{2}$
This relation is derived using paraxial rays.
It is true for both concave and convex spherical mirrors.
Only the sign of f and R changes with mirror type.
Hence it is valid for both.
4. **Correct answer is: A) $v = -15$ cm**
Using mirror formula:
$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$
$$\frac{1}{-10} = \frac{1}{v} + \frac{1}{-30}$$
Solving gives $v = -15$cm, so image is in front of the mirror.
5. **Correct answer is: A) 60 cm**
Substitute values:
$$\frac{1}{-15} = \frac{1}{v} + \frac{1}{-20}$$
$$\frac{1}{v} = -\frac{1}{15} + \frac{1}{20} = \frac{-4+3}{60} = -\frac{1}{60}$$
$v = -60$ cm
6. **Correct answer is: C) +13.3 cm**
Apply mirror formula:
$$\frac{1}{20} = \frac{1}{v} + \frac{1}{-40}$$
$$\frac{1}{v} = \frac{1}{20} + \frac{1}{40} = \frac{3}{40}$$
So $v = +13.3$cm, positive shows virtual image.

7. **Correct answer is: A) reciprocal of its focal length**
Power of an optical device is defined as the reciprocal of its focal length.
$$P = \frac{1}{f}$$
It tells how strongly the device converges or diverges light.

8. **Correct answer is: C) D**
The SI unit of power of a lens or mirror is dioptre (D).
It is used because power is the reciprocal of focal length in metres.
So the correct unit is D.

9. **Correct answer is: B) -5D**
Answer is b) Focal length, f=20 cm=0.2 m
Using the formula:
P=−1/f
P=−1/0.2
=−5 D
Answer: −5 D

10. **Correct answer is: C) -2 D**
Answer is C) Focal length, f=+0.5 m
Using the formula:
P=−1/f
P=−1/0.5
=−2 D
Answer: −2 D

Worksheet-4

1. **Correct option: (B)**
For a short object, longitudinal magnification is defined as
$m_L = \frac{dv}{du}$.
From mirror formula differentiation, we get
$\frac{dv}{du} = -\frac{v^2}{u^2}$.
So, $m_L = -\frac{v^2}{u^2}$.

2. **Correct option: (C)**
A long object has two ends at distances $u_1$and u_2.
Its length is the difference between these positions.
So, length = $u_1 - u_2$.

3. **Correct option: (B)**
Longitudinal magnification is image length divided by object length.
So, $m_L = -\frac{v_1 - v_2}{u_1 - u_2}$.
Due to sign convention for mirrors, it becomes negative.

4. **Correct option: (C)**
For a short object, longitudinal magnification is defined by
$m_L = \frac{dv}{du}$.
It relates the small change in image distance to object distance.

5. **Correct option: (C)**
Concave parabolic mirrors convert light from a source at focus into parallel rays.
This produces a strong, directed beam of light.
Hence they are used in headlights and searchlights.

6. **Correct option: (C)**
When the face is placed between pole and focus,
a concave mirror forms a virtual, erect and magnified image.
This helps in seeing facial details clearly.

7. **Correct option: (C)**
Convex mirrors give a wider field of view.
they provide a wider field of view and form diminished, erect images.

8. **Correct option: (B)**
A rear-view mirror must show a wide area behind the vehicle.
Convex mirrors provide a wider field of view.
Hence they are used as rear-view mirrors.

9. **Correct option: (C)**
To find the radius of curvature (R), we use the magnification formula
$m = \frac{f}{f-u}$, where $m = +4$(upright virtual image) and $u = -0.60$cm.
Plugging in the values:
$4 = \frac{f}{f-(-0.60)} \Rightarrow 4f + 2.4 = f \Rightarrow 3f =$

-2.4, giving a focal length $f = -0.80$cm.
The radius of curvature is $R = 2f$, so $R = 2 \times (-0.80\text{ cm}) = -1.60$ cm(or 1.60 cm in magnitude for a concave mirror).

10. Correct option: (C)
Using the mirror formula and differentiating, we get that image speed is proportional to $(v^2/u^2) \times$ object speed. Substituting v = 0.9 m, u = 10 m, and object speed = 5 m/s, we get image speed = 0.0405 m/s. Hence, the image moves much slower because $(v/u)^2$ is very small.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct answer: C)
Reflection of light is the bouncing back of light rays when they strike a smooth and shiny surface like a mirror. It allows us to see images.

2. Correct answer: C)
Only a part of the incident light is reflected by water while most of it goes inside (is transmitted), so the reflected image appears less bright.

3. Correct answer: A)
A concave mirror placed with a point source at its focus reflects light rays parallel to the principal axis, forming a parallel beam.

4. Correct answer: A)
A convex mirror always forms a virtual, erect image. It never forms an inverted image for any position of the object.

5. Correct answer: B)
A concave mirror can produce a virtual, erect and magnified image when the object is placed between the pole and the focus.

6. The correct option is D) Zero.
For a concave mirror, the minimum distance occurs when the real object is placed at the center of curvature (u = 2f). At this specific position, the real image is formed at the exact same location, resulting in a distance between them of zero.

7. Correct option: C
For a real image in a concave mirror, magnification $m = -\frac{v}{u} = -n$.
So $v = nu$. Using mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{nu} + \frac{1}{u} = \frac{n+1}{nu}$.
Hence $u = \frac{n+1}{n} f$.

8. Correct option: B
Distances from focus are $x_1 = u - f$and $x_2 = v - f$.
Using Newton's form of mirror formula, $uv = f(u + v)$gives $(u - f)(v - f) = f^2$.
So $x_1 x_2 = f^2 \Rightarrow f = \sqrt{x_1 x_2}$.

9. The correct option is D) It is a concave mirror of focal length 24 cm.
The mirror is concave because it forms an erect, enlarged image (m = +4). Using v = −4u and the distance |v − u| = 90 cm, we find u = −18 cm and v = 72 cm. Applying the mirror formula, the focal length is calculated as 24 cm

10. Correct option: B
Using the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$with a virtual object distance $u = +10$cm and focal length $f = +15$cm, the image distance $v$is calculated to be -30cm. A negative $v$indicates a real image formed in front of the mirror, and the magnification

$$m = -\frac{v}{u} = -\frac{-30}{10} = +3$$

confirms it is erect and enlarged three times. The correct answer is B) Real, erect, and enlarged three times.

11. Correct option: D
To find the distance of the mirror (u) from the candle, we use the mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

and the fact that the image is formed on a wall 1.5 m behind the candle $(v = u + 1.5)$.
Substituting $f = -1$m (for a concave mirror to form a real image) into

$$\frac{1}{-1} = \frac{1}{-(u + 1.5)} + \frac{1}{-u}$$

gives the quadratic equation

$$u^2 + 1.5u - 1.5 = 0.$$

12. Correct option: D

A concave mirror gives image twice the size of object at two positions: one between F and C (real) and one between P and F (virtual).

Since the object is shifted away from the mirror to get the second position, the first must be the nearer one.

Thus, the object was initially between P and F.

13. Correct answer: C) Real and erect

Using the mirror formula $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$with a virtual object distance $u=+20$cm and focal length $f=+30$cm, we find $v=-60$cm. Since the image distance $v$is negative, the image is formed in front of the mirror and is therefore **real**. The magnification $m=-\frac{v}{u}=-\frac{-60}{20}=+3$is positive, which means the image is **erect** (Option C).

14. Correct answer: B) 15 cm

Magnification $m=\frac{h_i}{h_o}=3=-\frac{v}{u}$(negative for real, inverted image).

Given $u=-20$cm, so $v=-60$cm.

Using mirror formula $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}=\frac{1}{-20}+\frac{1}{-60}=\frac{-4}{60}$,

so $f=-15$cm → magnitude $=15$cm.

15. Option C.

For a convex mirror, take $f=+10$cm and $u=-20$cm.

Using mirror formula $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{1}{10}=\frac{1}{v}-\frac{1}{20}$.

So $\frac{1}{v}=\frac{3}{20}\Rightarrow v=\frac{20}{3}$cm (positive → behind mirror).

Hence image is **virtual at $\frac{20}{3}$cm**

16. Option A.

Radius $R=15$cm ⇒ focal length $f=\frac{R}{2}=7.5$cm (concave ⇒ $f=-7.5$cm, $u=-10$cm).

$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow-\frac{1}{7.5}=\frac{1}{v}-\frac{1}{10}\Rightarrow v=-30$cm.

Magnification $m=-\frac{v}{u}=-\frac{-30}{-10}=-3$(real, inverted, 3 times).

17. Option D.

Magnification $m=\frac{v}{u}=-4.5$(negative since image is real and inverted), $u=-20$cm.

So $v=4.5\times20=90$cm (negative sign cancels in calculation).

Using $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}=\frac{1}{-90}+\frac{1}{-20}=-\frac{11}{180}$.

Thus $f=\frac{180}{11}$cm

18. Correct option: C) A convex mirror

A convex mirror always forms a virtual and erect image.

The image is always smaller (diminished) than the object.

Hence only a convex mirror gives a diminished virtual image.

19. Correct option: A) 75 cm

For doubled and inverted image, magnification $m=-2$.

Using $m=-v/u$, we get $v=2u$.

Substituting in mirror formula gives $u=75$cm.

20. Correct option: A) 2 cm

At $u=4f$, image is formed at $v=\frac{4f}{3}$.

Magnification $m=\frac{v}{u}=\frac{1}{3}$.

Image length $=\frac{1}{3}\times6=2$cm.

21. Correct option: D) 15 cm behind the mirror

The image forms **15 cm behind the mirror**, exactly midway between the pole and the principal focus.

Since the image distance *(v)* is positive, the image is **virtual, erect, and diminished** in size.

This is calculated using the mirror formula

$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$, where $f=+30$cm and $u=-30$cm

22. The correct answer is D) Always.

A convex mirror always forms a virtual, erect, and diminished image of a real object, regardless of the object's distance *(u)* from the mirror. This is because the reflected rays always diverge and appear to meet at a point behind the mirror between the pole and the focus

23. Correct option: A

The object is at 40 cm which is 2f (since f = 20 cm).
For a concave mirror, an object at 2f forms image at 2f.
Hence image is real, inverted and same size.

24. Correct option: D

A convex mirror always forms a virtual, erect and diminished image.
So A, B and C are correct properties.
The wrong statement is image is real.

25. Correct option: A

Given f = +10 cm and magnification m = +1/2 (since the image is virtual and erect), we have $v = -u/2$. Applying the mirror formula $1/f = 1/v + 1/u$, we get $1/10 = -2/u + 1/u$. Solving this yields $1/10 = -1/u$, which results in $u = -10$ cm. The object is placed at a distance of **10 cm** in front of the convex mirror.

26. Answer: A) 8 cm

Magnification $m = \frac{v}{u} = 16$.
The image distance (wall) $v = 120$cm, so $u = \frac{120}{16} = 7.5$cm ≈ 8 cm.
Hence the flower should be placed 8 cm from the mirror.

27. Correct option: C

For image twice the object, magnification $m = 2$.
Using mirror relations for a concave mirror, object and image distances differ accordingly.
The separation comes out to be 30 cm.

28. The correct answer is B) Convex.

A convex mirror always forms a virtual, erect, and diminished (smaller) image regardless of the object's distance.

Since the man's height (6 m) is reduced to an image height of 2 m while remaining erect, it fits the property of a convex mirror.

A plane mirror would show a 6 m image, and a concave mirror would only show an erect image if it were magnified (larger than 6 m).

29. Correct option: C) It minimises spherical aberration and provides a parallel beam of light.

A parabolic reflector sends light from the focus into a parallel beam.
It reduces spherical aberration, giving a sharp and bright beam.
Hence automobile headlights use parabolic reflectors, option C.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

30. Correct statements – B, C, D

A convex mirror always forms a virtual image, never a real one.
A concave mirror can produce a magnified image (object between F and C).
A concave mirror can also form a real and inverted image.

31. Correct options – A and B

A plane mirror always gives a virtual and erect image.
A convex mirror also always forms a virtual and erect image for any object position.
A concave mirror does not always do so.

32. Correct statements – A and B

A concave mirror forms a real image when the object is beyond the focus.
It forms a virtual image when the object is between pole and focus.
Hence both types are possible.

33. Correct options – A and B

Magnification $m = 2$implies $\frac{v}{u} = 2$and using mirror formula.
Solving gives two object positions: one between F and C and one between P and F.
These correspond to $u = \frac{f}{2}$and $u = \frac{3f}{2}$.

34. Correct options – A, C and D

Pole, radius of curvature and principal axis are geometrical properties of the mirror.
They do not depend on whether rays are paraxial or marginal.

The focus depends on paraxial approximation, so it changes.

ASSERTION AND REASON TYPE

35. Correct Answer: (a)
A convex mirror gives a wide field of view.
It always forms virtual, erect and diminished images.
This helps the driver see more area behind, so it is correctly used as a driver's mirror.

36. Correct Answer: (c)
A virtual image cannot be obtained on a screen, so the Assertion is true.
But virtual images are **not** formed by actual meeting of rays.
They are formed by the apparent meeting of rays, so the Reason is false.

37. Correct Answer: (a)
Spherical mirrors have curved reflecting surfaces.
They are actually parts of a sphere.
Hence, the Reason correctly explains the Assertion.

38. Correct Answer: (a)
Concave mirrors are used in searchlights and headlights.
Concave parabolic mirrors focus light into a strong parallel beam.
This strong beam helps illuminate long distances, so the Reason explains the Assertion.

COMPREHENSION TYPE

PASSAGE-I

39. Correct option: A
Using mirror formula $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$.
$\frac{1}{-20}=\frac{1}{v}+\frac{1}{-30}\Rightarrow\frac{1}{v}=-\frac{1}{60}\Rightarrow v=-60\text{cm}$.
Negative sign shows image is formed 60 cm in front of the mirror.

40. Correct option: B
Magnification $m=-\frac{v}{u}$.
$m=-\frac{-60}{-30}=2$.
So the image is twice the size of the object.

41. Correct option: A
New object distance $u=-20\text{cm}$ which equals focal length $f=-20\text{cm}$.
For object at focus, image is formed at infinity.
Hence image moves to infinity.

PASSAGE- II

42. Correct option: B) $\frac{20}{3}$cm behind the mirror
Using mirror formula $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$.
$\frac{1}{10}=\frac{1}{v}-\frac{1}{20}\Rightarrow\frac{1}{v}=\frac{3}{20}\Rightarrow v=\frac{20}{3}\text{cm}$.
Positive $v$means image is formed behind the mirror (virtual).

43. Correct option: A) $\frac{1}{3}$
Magnification $m=-\frac{v}{u}$.
$m=-\frac{20/3}{-20}=\frac{1}{3}$.
So image is diminished and erect.

44. Correct option: C) 0.167 cm
$h_i=m\times h_o=\frac{1}{3}\times 0.5$.
$h_i=0.1667\approx 0.167\text{cm}$.
Hence the image is smaller than the object.

MATRIX MATCHING TYPE

45. a – Q, b – S, c – P, d – T

a) Object at focus of a convex mirror → T (m = 0.33)
A convex mirror always forms a virtual, erect and diminished image.
At F, the image lies between F and P, giving strong diminution.
So magnification is less than 1, nearly 0.33.

b) Object at centre of curvature of a concave mirror → S (m = –1)
At C, a concave mirror forms an image at C itself.
The image is real, inverted and same size as the object.
Hence magnification = –1.

c) Object at focus of a concave mirror → P (m = –∞)
At focus, reflected rays become parallel.
Image is formed at infinity and is highly enlarged.
Thus magnification is –∞.

d) Object at radius of curvature of a convex mirror → Q (m = 0.5)
A convex mirror always gives a diminished image.

At large distances, image comes closer to focus.
Hence magnification is about 0.5.

46. a – Q, T

b – P, S
c – P, T
d – R, T

a) m = 1/4 → Q, T (Convex mirror, Virtual)
Positive and less than 1 means virtual and diminished.
Only a convex mirror forms such images.
So it is convex and virtual.

b) m = –1 → P, S (Concave mirror, Real)
Negative magnification means image is inverted.
Inverted images are always real.
Only concave mirror can give this case.

c) m = 2 → P, T (Concave mirror, Virtual)
Positive and greater than 1 means virtual and magnified.
This happens when object is between P and F of concave mirror.
So it is concave and virtual.

d) m = 1 → R, T (Plane mirror, Virtual)
Magnification of plane mirror is always +1.
Image is same size and erect.
Hence it is plane mirror and virtual.

47. The lateral magnification of the object is 2

Given that $u = -5$cm and $f = -10$cm for a concave mirror, we use the magnification formula
$m = \frac{f}{f-u}$. By substituting the values,
$m = \frac{-10}{-10-(-5)} = \frac{-10}{-5} = 2.$, indicating a virtual and erect image.

48. Given $f = -10$cm and image is twice ⇒ $m = -2$(real, inverted).

So $v = 2u$; using mirror formula gives $u = -15$cm and $v = -30$cm.
Distance between object and image $=| v - u | = 15$cm.

49. For convex mirror $f = +20$cm $= 0.20$m.

Power $P = \frac{1}{f} = \frac{1}{0.20} = +5$D.
Positive sign shows diverging (convex) mirror.

50. Lateral magnification $m = -\frac{v}{u}$.

Given $v = 4u$, so $m = -\frac{4u}{u} = -4$.
Negative sign shows image is inverted.

PREVIOUS YEAR QUESTIONS

1. Correct option: B)

Find Initial Position: Using the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$with $u = -19\ m$and $f = +1\ m$, the initial image position is

$$v_1 = \frac{19}{20} = 0.95\ m.$$

Find Final Position: After 1 second, the object moves 4 m closer ($u = -15\ m$), which results in a new image position of

$$v_2 = \frac{15}{16} = 0.9375\ m.$$

Calculate Speed: The average speed is the distance moved by the image divided by time:
$| 0.95 - 0.9375 |/1\ s = 0.0125\ m/s$(or $\frac{1}{80}\ m/s$).

2. Correct option: A)

Initially, the man is at distance *y*; after moving 5 m away, his new distance from the mirror is *y* + *5*.

The total distance between the man and his image is twice his distance from the mirror, so *2(y + 5) = 20*.

Dividing by 2 gives *y + 5 = 10*, which results in the initial distance *y = 5 metres*.

3. Correct option: B)

Case 1: With $m = -6$and object distance $u = -x$, the formula

$$-6 = \frac{-F}{-F-(-x)}$$

simplifies to the relation $7F = 6x$.

Case 2: When moved to $u = -(x+3)$with $m = -4$, the formula

$$-4 = \frac{-F}{-F+x+3}$$

simplifies to the equation $5F - 12 = 4x$.

Final Step: Substituting $x = \frac{7F}{6}$from the first equation into the second gives

$$5F - 12 = \frac{14F}{3},$$

which solves to $F = 36$ cm.

4. Correct option: C)

When a plane mirror rotates by an angle θ, the reflected ray rotates by double that angle (2θ). Since the mirror rotates by 15°, the reflected ray shifts by an angle of 2 × 15° = 30°. The displacement d on a screen at distance L is found using the trigonometric formula d = L tan (2θ). Substituting L = 2.0 m and 2θ = 30°, the displacement is d = 2 × tan (30°) ≈ 1.154 m.

5. Correct option: C)

Statement 1 is true, as convex mirrors always form upright *(positive m)* virtual images, while concave mirrors form both upright virtual and inverted *(negative m)* real images.

Statement 2 is false, because magnification is defined as the ratio of image height to object height *($m = h_i / h_o$)*, and its sign is an absolute physical property indicating orientation, not a choice of convention.

6. Correct option: C)

For a convex mirror, the magnification is

$$m = \frac{1}{2N+1}$$

and the focal length is $f = \frac{R}{2}$.

Using the relationship $m = \frac{f-v}{f}$,

we substitute the values to get $\frac{1}{2N+1} = \frac{\frac{R}{2}-v}{\frac{R}{2}}$,

which simplifies to

$$1 - \frac{2v}{R} = \frac{1}{2N+1}.$$

Solving for $v$gives

$$\frac{2v}{R} = 1 - \frac{1}{2N+1} = \frac{2N}{2N+1},$$

resulting in the image distance

$$v = \frac{NR}{2N+1}.$$

7. Correct option: D)

For a short linear object lying along the principal axis, the **longitudinal magnification** is calculated as $m_L = \left(\frac{f}{f-u}\right)^2$.

Substituting the given values ($f = -15$ cm and $u = -26.5$cm), the magnification is approximately $\left(\frac{-15}{-15+26.5}\right)^2 = \left(\frac{-15}{11.5}\right)^2 \approx 1.7$.

Multiplying the object length by this factor (3 cm × 1.7), the size of the image is approximately **5.1 cm**.

8. The correct answer is B.

Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.

Statement 1 is True: We can identify a mirror by the size of the erect image: a Plane mirror shows the same size, a Convex mirror shows a diminished image, and a Concave mirror shows a magnified image.

Statement 2 is True: For a concave mirror, when a real object is placed at the Center of Curvature (C), the real image is formed at the exact same location, making the distance between them zero.

Relationship: Statement 2 describes a specific property of real images in concave mirrors, which does not explain the logic used in Statement 1 to identify mirrors using erect (virtual) images

2. Light- Refraction

Solutions

Worksheet – 1

1. **Correct Answer : D**
   The angle of refraction (r) is strictly defined as the angle between the refracted ray (the light path inside the material) and the normal (the imaginary line perpendicular to the surface at the point of entry).
2. **Correct Answer :C optically rarer**
   An optically rarer medium has lower refractive index.
   Light travels faster in a medium of lower refractive index.
   Hence, such a medium is called optically rarer.
3. **Correct Answer : B slows down and bends towards the normal**
   A denser medium has a higher refractive index.
   Light slows down when entering a denser medium.
   Because of this decrease in speed, the ray bends towards the normal.
4. **Correct Answer : C refractive index of second medium with respect to the first**
   Snell's law is written as μ = sin i / sin r.
   Here μ measures how much light bends between two media.
   Thus μ is the refractive index of the second medium w.r.t the first.
5. **Correct Answer :B sin i / sin r**
   Snell's law states sin i / sin r = constant.
   This constant is the refractive index of the two media.
   Hence sin i / sin r remains constant.
6. **Correct Answer : C Angle of incidence**
   The incident ray strikes the surface at a point.
   The normal is the perpendicular line at that point.
   The angle between them is called the angle of incidence.
7. **Correct Answer: C 2**
   Refractive index $\mu = \frac{\lambda_{vacuum}}{\lambda_{medium}}$.
   $\mu = \frac{4\times10^{-7}}{2\times10^{-7}} = 2$.
   So, the refractive index of the medium is 2.
8. **Correct Answer: B 2×10^8 m/s**
   Refractive index $\boldsymbol{\mu = \frac{c}{v}}$.
   So $\boldsymbol{v = \frac{c}{\mu} = \frac{3\times10^8}{1.5}}$.
   $\boldsymbol{v = 2 \times 10^8}$ m/s.
9. **Correct Answer: C frequency**
   Refractive index depends on speed and wavelength of light.
   The frequency of light remains constant when light changes medium.
   Hence, refractive index is independent of frequency.
10. **Correct Answer: C**
    Since $\mu_{21} = \frac{v_1}{v_2}$.
    Given $v_1 > v_2$, so $\frac{v_1}{v_2} > 1$.
    Therefore, the relative refractive index μ_{21} is greater than 1

Worksheet-2

1. **Correct Answer: B Only twice**
   A ray refracts once when it enters the glass slab.
   It refracts again when it comes out of the slab.
   Hence, total refractions are two.
2. **Correct Answer: B μg / μ_a = sin i / sin r**
   Snell's law gives refractive index as sin i divided by sin r.
   Here light goes from air to glass.
   So refractive index of glass w.r.t air is μg / μ_a = sin i / sin r.
3. **Correct Answer: B The angle of refraction at first surface (r)**
   Inside the slab, the ray travels in a straight line.
   So the angle it makes with the normal remains unchanged.
   Hence r′ = r.

4. **Correct Answer: C) Retraces its original path**
The principle of reversibility states light follows the same path.
If the direction is reversed, the ray returns along the same route.
Thus it retraces its original path.
5. **Correct Answer: C h ÷ μ**
Apparent depth is less than real depth in a denser medium.
It is given by real depth divided by refractive index.
So apparent depth = h / μ.
6. **Correct Answer: B Apparent depth = Real depth ÷ μ**
Light bends when going from water to air.
This makes the object appear closer than it really is.
Hence apparent depth = real depth ÷ μ.
7. **Correct Answer: B**
Snell's law for refraction is $\mu_1 \sin\ \theta_1 = \mu_2 \sin\ \theta_2$.
Here θ_1 is the angle in medium μ_1 and θ_2 in μ_2.
8. **Correct Answer: D**
Speed of light $v = \frac{c}{n}$, so it is maximum in the medium with least refractive index.
From the figure, the ray bends most away from the normal in medium n_4, indicating lowest n.
Hence, velocity is maximum in n_4
9. **Correct Answer: C — Principle of reversibility of light**
When a mirror is placed, the emergent ray reflects back.
It travels along the same path in the opposite direction.
10. **Correct Answer: C — Not be parallel to the incident ray**
Parallelism occurs only when the surrounding media are the same.
Different media change the refraction at each surface.
So the emergent ray is not parallel to the incident ray.

Worksheet-3

1. **Correct answer: B — Ratio of image size to object size**
Magnification tells how many times bigger or smaller the image is compared to the object.
It is defined as image height divided by object height.
2. **Correct answer: C — Perpendicular to the principal axis**
Lateral magnification is used when the object stands upright across the principal axis.
Both object and image are measured in height perpendicular to the axis.
3. **Correct option: B**
$$m = \frac{I}{O}$$
(Lateral magnification = image size / object size.)
4. **Correct option: B** $\boldsymbol{P = \frac{\mu_2 - \mu_1}{R}}$
(Power depends on the change in refractive index across the surface.)
5. **Correct option: B a – Biconcave, b – Plano-concave, c – Concavo-convex**
6. **Correct answer: B Convex and concave**
Lenses are mainly classified based on their shape and how they bend light.
Convex lenses converge light rays, while concave lenses diverge them.
7. **Correct answer: C) Converging**
A convex lens brings parallel rays of light together after refraction.
Hence, in air it is called a **converging lens**.
8. **Correct answer: C) 2 D**
Refracting power P is the inverse of reduced focal length.
$$P = \frac{1}{(f/\mu)} = \frac{1}{0.50} = 2\ \text{D}$$
9. **Correct answer: B) Reduced focal length**
Refracting power of a spherical surface is defined as
$$P = \frac{1}{(f/\mu)}$$
so it is the inverse of the reduced focal length.

10. **Correct answer: C) Length of image / length of object**
Longitudinal magnification compares the size of image and object along the principal axis, i.e.
Longitudinal magnification $= \frac{\text{length of image}}{\text{length of object}}$.

Worksheet-4

1. **Centre of curvature → C**
It is the centre of the imaginary sphere of which the lens surface is a part.
Each curved surface of a lens has its own centre of curvature.
2. **Convex lens & parallel rays → C**
A convex lens is a converging lens.
Parallel rays after refraction meet at a point on the principal axis called the principal focus.
3. **Aperture → B**
The aperture is the effective diameter of the lens.
It determines how much light can pass through the lens.
4. **Object at infinity (convex lens) → B**
When the object is at infinity, rays are parallel.
A convex lens brings parallel rays to focus at its focal point (20 cm).
5. **f = 15 cm, u = 30 cm (convex lens) → A**
Correct answer: C) At focus F_2
Using the lens formula:
1/f = 1/v + 1/u
Since u = ∞, 1/u = 0, so 1/v = 1/f ⇒ v = f
Therefore, the image is formed at the focus (F_2).
6. **Virtual image by convex lens → C**
Answer: B) 30 cm
Given: f = +15 cm, u = -30 cm
Using lens formula:
1/f = 1/v - 1/u
1/15 = 1/v + 1/30
1/v = 1/15 - 1/30 = 1/30
v = 30 cm
7. **Virtual image by convex lens → C**
Answer: C) Between focus and optical centre
A convex lens forms a virtual image when the object is placed between the focus and the optical centre.
8. **Concave lens, object at infinity → B**
A concave lens always forms a virtual and erect image.
For an object at infinity, the image is formed at the focus.
9. **Concave lens: f = −15 cm, u = −30 cm**
Using lens formula:
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
$\frac{1}{-15} = \frac{1}{v} + \frac{1}{-30}$
$\frac{1}{v} = -\frac{1}{15} + \frac{1}{30} = -\frac{1}{30} \Rightarrow v = -30$ cm
The negative sign shows the image is virtual and on the same side as the object.
10. **Correct answer: A) Positive and less than 1**
A concave lens always forms a virtual and erect image, so magnification is positive. The image is always smaller than the object, so the magnification is less than 1.

Worksheet-5

1. **Correct Answer: Option B**
The correct lens formula is $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$.
Using sign convention, it is written as $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$.
2. **Correct Answer: A) Positive and less than 1**
A concave lens always forms a virtual and erect image, so magnification is positive. The image is always smaller than the object, so the magnification is less than 1.
3. **Correct Answer: A) Positive and less than 1**
A concave lens always forms a virtual and erect image, so magnification is positive. The image is always smaller than the object, so the magnification is less than 1.

4. **Correct Answer: C) greater than four times the focal length**

In the displacement method, two sharp images are obtained only when the distance between object and screen is greater than 4f.

5. **Correct Answer: D) concave lens**

A concave lens always forms a virtual image which cannot be obtained on a screen, so displacement method cannot be used.

6. **Correct Answer: C) Imaginary**

When d < 4f, the value under the square root becomes negative, so u becomes imaginary and no real image is possible.

7. **Correct Answer: C) Ratio of image height to object height**

Lateral magnification tells how much the size of image changes, so it is the ratio of image height to object height.

8. **Correct Answer: B) m = v / u**

For a lens, magnification is defined as m = image distance (v) ÷ object distance (u).

9. **Correct Answer: B) dioptre**

Power of a lens is 1/f (in metres) and its SI unit is dioptre (D).

10. **Correct Answer: C) positive**

A convex lens converges light rays, so its focal length and power are positive.

Worksheet-6

1. **Correct Answer: A**

Total magnifying power $M = m_o \times m_e$.
Given $M = 100$ and $m_e = 4$.
So $m_o = 100/4 = 25$.

2. **Correct Answer: C**

From denser to rarer medium, light bends away from the normal.
Hence angle of refraction becomes greater than angle of incidence.

3. **Correct Answer: D**

Critical angle is the angle of incidence for which refracted ray makes 90°. Beyond this, total internal reflection occurs.

4. **Correct Answer: B**

Magnifying power of a simple microscope is $m = \frac{D}{f}$ (or D/u when object at near point).
So correct form is $m = D/u$.

5. **Correct Answer: C**

Optical instruments use reflection in mirrors and refraction in lenses.
Hence both laws are used.

6. **Correct Answer: C**

A simple microscope uses one convex lens of short focal length.
It acts as a magnifying glass.

7. **Correct Answer: B**

The objective lens must form a large magnified image, so it has small focal length and small aperture.

8. **Correct Answer: C**

In a compound microscope, the objective forms its image beyond $2F_0$.
This image is real, inverted and magnified.

9. **Correct Answer: C**

Magnifying power is defined as the ratio of angle subtended by final image to angle subtended by object.

10. **Correct Answer: C**

Total magnification equals product of objective and eyepiece magnifications.
So $m = m_o \times m_e$.

Competitive Corner

SINGLE CORRECT ANSWER TYPE

1. Correct Answer: A

In a convex lens, the middle part is thicker, so light bends more there.
Greater bending means higher refractive index.
Hence μ (centre) > μ_2 (middle) > μ_1 (edge)

2. Correct Answer: C

A real image formed by a convex lens is inverted (both top–bottom and left–right).
So the given object rotates by 180° in the image.
Thus 1↔3 and 2↔4,

3. Correct Answer: A

For refraction at a spherical surface:

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Here, $\mu_1 = 1.5$(glass), $\mu_2 = 1$(air), $u = -3$cm, $R = +5$cm.
Solving gives $v = -2.5$cm, so image appears 2.5 cm from the surface

4. Correct Answer: B

Power of combination:

$$P = P_1 + P_2 - \frac{d}{f_1 f_2}$$

For zero power, $P = 0$.
Hence $d = f_1 + f_2$.

5. Correct Answer: (C).

Magnification:

$$\mathrm{m} = \frac{\mathrm{v}}{\mathrm{u}} = \frac{1}{\mathrm{n}}$$

Using lens formula and magnification relation,

$$\mathrm{u} = (\mathrm{n} + 1)\mathrm{f}$$

6. Correct Answer: A

Magnification:

$$\mathrm{m} = \frac{\mathrm{v}}{\mathrm{u}}$$

Using lens formula $\frac{1}{\mathrm{f}} = \frac{1}{\mathrm{v}} + \frac{1}{\mathrm{u}}$ and $\mathrm{v} = \mathrm{mu}$,

$$u = \frac{(m+1)f}{m}$$

7. Correct Answer: D

The convex lens is divided into three parts made of different materials, so each part has a different refractive index and hence a different focal length.
Each part forms its own image of the same point object on the axis.
Therefore, 3 images are formed

8. Correct Answer: D)

The object is 15 cm in front of a convex spherical glass surface of radius R = 30 cm. Using refraction at a spherical surface, the image is formed on the same side (air side) at 18 cm from the surface.
So the image is 18 cm to the left

9. Correct Answer: B

Even if the central part of a convex lens is covered, every small part of the lens can form a complete image. Covering part only reduces the amount of light passing through, so the image becomes dimmer but complete.

10. Correct Answer: C

For two positions of a lens, the magnifications are inversely related, so $\frac{h_1}{h_2} = \frac{m_1}{m_2} = \frac{u_2}{u_1}$ and the object height remains the same.
Using $h_1 = 8$cm and $h_2 = 2$cm, object height $= \sqrt{8 \times 2} = 4$cm.

11. Correct Answer: C

First find image distance by objective using lens formula, then use eyepiece formula for final image at 20 cm.
Adding both image distances gives the separation between lenses, which comes out to 11 cm.

12. Correct Answer: B

Power of a lens in a medium is reduced because refraction depends on relative refractive index.

$$P_{\text{medium}} = P_{\text{air}} \frac{(\mu_l/\mu_m - 1)}{(\mu_l - 1)}.$$

So the power becomes smaller in water → **≈ 2.0 D**

13. Correct Answer: B

For normal adjustment,

$$M = \frac{L}{f_o} \times \frac{D}{f_e}$$

where $L = 20$cm, $f_o = 1$cm, $f_e = 2$cm, $D = 25$cm.

$$M = \frac{20}{1} \times \frac{25}{2} = 250$$

14. Correct Answer: D

Magnifying power of compound microscope is

$$\mathrm{m} = \frac{-\mathrm{v_o}}{\mathrm{u_o}}\left(1+\frac{\mathrm{D}}{\mathrm{f_e}}\right)$$

15. Correct Answer: B

Net power $P = P_1 + P_2 = +6 + (-4) = +2\,D$.

Positive power means **convex lens**.

$$f = \frac{1}{P} = \frac{1}{2} = 0.5\,m = 50\,cm$$

16. Correct Answer: A

Let distance from S_1be x, so from S_2it is $24 - x$.

For images at same place:

$$\frac{1}{9} = \frac{1}{x} + \frac{1}{24-x}$$

Solving gives $x = 6$cm or 18cm (only 6 cm fits between).

17. Correct Answer: B

When a convex lens is cut (horizontally or vertically), only its **aperture** changes, not the **curvature of surfaces**.

Since focal length depends only on curvature and refractive index, part **A** has the **same focal length** as the original lens.

18. Correct Answer: C

For a simple microscope forming image at near point $D = 25$cm,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Here $f = 2.5$cm, $v = -25$cm (virtual image).

$$\frac{1}{2.5} = \frac{1}{-25} + \frac{1}{u} \Rightarrow \frac{1}{u} = 0.4 + 0.04 = 0.44$$
$$\Rightarrow u \approx 2.27 \text{ cm}$$

19. Correct Answer: B

For refraction at a spherical surface,

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Here image at infinity $\Rightarrow v = \infty$, $\mu_1 = \mu(\text{slab})$, $\mu_2 = 1(\text{air})$.

$$0 - \frac{\mu}{u} = \frac{1-\mu}{R} \Rightarrow u = \frac{\mu R}{\mu - 1}$$

20. Correct Answer: C

Relation between $\boldsymbol{n_1}$and $\boldsymbol{n_2}$

The lens shown is convex and is causing the parallel rays to converge. A convex lens converges light only when the lens material has higher refractive index than the surrounding medium.

$$\boxed{n_1 > n_2}$$

21. Correct Answer: D)

Power of two thin lenses in contact

For lenses in contact, powers add:

$$P = P_1 + P_2 = \frac{1}{f_1} + \frac{1}{f_2}$$

This corresponds to

$$P = \frac{f_1 + f_2}{f_1 f_2}$$

22. Correct Answer: B

Power of a spherical refracting surface

Power of a refracting surface:

$$P = \frac{(n_2 - n_1)}{R}$$

$n_1 = 1$, $n_2 = \frac{4}{3}$, $R = 10$ cm $= 0.1$ m

$$P = \frac{\frac{4}{3} - 1}{0.1} = \frac{\frac{1}{3}}{0.1} = 3.33\,D$$

23. Correct Answer: A

Answer is b)

Use lens maker formula: 1/f = (μ − 1)(1/R_1 − 1/R_2)

Substitute values:

1/10 = 0.5(1/7.5 − 1/R_2)

Simplify:

1/5 = 1/7.5 − 1/R_2

⇒ 1/R_2 = −0.0667

⇒ R_2 = −15 cm

Final Answer: −15 cm

24. Correct Answer: A

$$P_1 = \frac{1}{0.5} = +2D, P_2 = \frac{1}{-1} = -1D$$
$$P = 2 - 1 = 1D$$

25. Correct Answer: B

For a simple microscope with final image at the near point (D = 25 cm), magnifying power is

$M = 1 + \frac{D}{f}$

26. Correct Answer: D

Answer is A)

$M = (v_0 / u_0) \times (D / f_e)$

$25 = (v_0 / u_0) \times (25 / 5)$

$\Rightarrow v_0 / u_0 = 5$

$L = v_0 + f_e$

$14 = v_0 + 5$

$\Rightarrow v_0 = 9$ cm

$u_0 = v_0 / 5 = 9 / 5 = 1.8$ cm

Answer: 1.8 cm

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

27. Correct Answer: C,D

For refraction at a spherical surface,

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

- First focus $f_1 \rightarrow$ image at infinity $(v = \infty)$

$$-\frac{\mu_1}{f_1} = \frac{\mu_2 - \mu_1}{R} \Rightarrow f_1 = \frac{\mu_1 R}{\mu_2 - \mu_1}$$

- Second focus $f_2 \rightarrow$ object at infinity $(u = \infty)$

$$\frac{\mu_2}{f_2} = \frac{\mu_2 - \mu_1}{R} \Rightarrow f_2 = \frac{\mu_2 R}{\mu_2 - \mu_1}$$

28. Correct Answer: A,C

For a plane surface, radius $R = \infty$.

So refraction formula becomes

$$\frac{\mu_2}{v} = \frac{\mu_1}{u}$$

Rearranging,

$$\frac{v}{u} = \frac{\mu_2}{\mu_1} \text{and} v\mu_1 = u\mu_2$$

29. Correct Answer: A,D

For refraction at a **spherical surface**, we use

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Here,

Object at infinity $\Rightarrow u = \infty \Rightarrow \frac{\mu_1}{u} = 0$

$\mu_1 = 1$(air), $\mu_2 = 1.5$, $R = +20$ cm(convex surface)

$$\frac{1.5}{v} = \frac{1.5 - 1}{20} = \frac{0.5}{20}$$
$$v = \frac{1.5 \times 20}{0.5} = 60 \text{ cm}$$

Since $v$is **positive**, rays **actually meet** inside the medium.

30. Correct Answer: B,C,D

When light enters a medium of refractive index μ:

- Speed changes: $v = \frac{v_0}{\mu}$
- Frequency remains unchanged: $f = f_0$
- Wavelength changes because $\lambda = \frac{v}{f} = \frac{v_0/\mu}{f_0} = \frac{\lambda_0}{\mu}$

31. Correct Answer: A,C

For a lens in a medium, focal length changes as

$$f' = f \times \frac{(\mu_g - 1)}{(\mu_g/\mu_m - 1)}$$

Here

Glass $\mu_g = \frac{3}{2}$, Water $\mu_m = \frac{4}{3}$

$$\mu_{g\ in\ water} = \frac{\mu_g}{\mu_m} = \frac{3/2}{4/3} = \frac{9}{8}$$
$$f' = f \times \frac{(3/2 - 1)}{(9/8 - 1)} = f \times \frac{1/2}{1/8} = 4f$$

So focal length **increases** to $4f \Rightarrow$ lens becomes **weaker**.

Originally the image was **real, inverted and magnified** $\Rightarrow$ object was between $f$and $2f$.

Now new focal length is $4f$, so the same object lies **within the focal length**, giving a **virtual, magnified image**.

ASSERTION AND REASON TYPE

32. Correct Answer: a

A ray bends when it enters a new medium because its speed changes. Different speeds in different media cause refraction, so R explains A.

33. Correct Answer: a

In a rectangular glass slab, the two faces are parallel, so the angle at the second surface equals the first refraction angle. Thus R correctly explains A.

34. Correct Answer: b

A ray through the optical centre passes undeviated — this is true.
But it is not because it lies on the principal axis; it is due to equal refractions at both surfaces canceling out.

35. Correct Answer: b

Ray diagrams do determine the position and nature of images.
But saying we can also calculate does not explain why ray diagrams work.

MARTIX MATCHING TYPE

36. Correct Answer: A – Q, B – Q, C – R, D – S

A. A convex lens in a denser medium → Q (Diverging lens)
When the surrounding medium is optically denser than the lens, the convex lens loses its converging power and behaves like a diverging (concave) lens.

B. A concave lens in a rarer medium → Q (Diverging lens)
A concave lens always spreads light rays. In a rarer medium it continues to behave as a diverging lens.

C. A plano-convex lens silvered on its curved surface → R (Concave mirror)
Light enters through the plane surface and is reflected from the silvered curved surface, which acts like a concave reflecting surface.

D. A plano-concave lens silvered on its plane surface → S (Convex mirror)
Light enters through the curved surface and reflects from the silvered plane surface, making it behave like a convex mirror.

37. Correct Answer: A – Q, B – R, C – S, D – P

A. Simple microscope → Q (Acts as magnifying glass):
A simple microscope is just a single convex lens used to magnify small objects.

B. Objective lens → R (Has small focal length):
The objective lens must have a small focal length to produce a highly magnified real image.

C. Eyepiece → S (Behaves like a simple microscope):
The eyepiece magnifies the image formed by the objective just like a simple microscope.

D. Compound microscope → P (Uses two converging lenses):
A compound microscope uses an objective and an eyepiece, both of which are converging lenses.

COMPREHENSION TYPE

Passage-I

38. Correct answer: B Virtual and erect at F_1

When the object is at infinity, rays falling on a concave lens are parallel. After refraction they diverge and appear to come from the focus **F_1**. Hence the image is **virtual, erect and highly diminished at F_1**.

39. Correct answer: C Virtual, erect and diminished

A concave lens always diverges light rays. So the image is always formed between F_1 and O, and is virtual, erect and smaller than the object.

40. Correct answer: C Beyond $2F_2$, enlarged and real

For a convex lens, when the object lies between F_1 and $2F_1$, the refracted rays meet beyond $2F_2$. Therefore the image formed is real, inverted and enlarged.

Passage-II

41. Correct answer: C

Region (A) is beyond the focal point on the left of a convex lens.

For a convex lens, when the object is beyond F, the image forms on the opposite side between F and 2F, i.e., region C.

42. Correct Answer: D

Converging rays mean the object is virtual and on the right side of the lens. A convex lens further converges them, so the image forms farther to the right, i.e., in region D.

43. Correct Answer: C

Using lens combination formula:

$$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}$$

$$\frac{1}{F}=\frac{1}{f}+\frac{1}{-3f}=\frac{3-1}{3f}=\frac{2}{3f}$$

$$F=\frac{3f}{2}$$

Since the focal length is positive, the system behaves like a convex lens, so the focus lies in region C.

INTEGER TYPE

44. Correct Answer: Total Magnifying Power (m) = m. × me

Substituting the given values:

m = 15 * 4

m = 60

45. Correct Answer: 1D

Power $P=\frac{1}{f}$.

For convex lens, $P_1=\frac{1}{0.5}=+2D$; for concave lens, $P_2=\frac{1}{-1}=-1D$.

Total power $P=P_1+P_2=2-1=1D$.

46. Correct Answer: 8.67

Given:

$f_o=2\text{cm}$, $f_e=4\text{cm}$, $u_o=-3\text{cm}$, $v_e=-8\text{cm}$

Objective lens:

$$\frac{1}{v_o}-\frac{1}{u_o}=\frac{1}{f_o}$$

$\Rightarrow v_o=6\text{cm}$

Eyepiece:

$$\frac{1}{v_e}-\frac{1}{u_e}=\frac{1}{f_e}$$

$\Rightarrow u_e=-\frac{8}{3}\text{cm}$

Distance between lenses:

$$L=v_o+|u_e|=6+\frac{8}{3}=\frac{26}{3}$$

$\boxed{8.67\text{ cm}}$

PREVIOUS YEAR QUESTIONS

1.Correct answer: A

The magnification is given by

$$m=\frac{f}{f+u}$$

I: $u=-16$ cm

$$m=\frac{f}{f-16}$$

II: $u=-8$ cm , $m=\frac{-f}{f-8}$

From (i) and (ii)

$$2\frac{f}{f-16}=\frac{-f}{f-8}$$

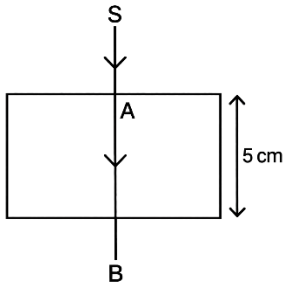
$$-(f-8)=f-16$$

$$2f=24\text{ cm}$$

$$f=12\text{ cm}$$

2. Correct answer: B

Let the distance from S to A is d.



Speed of light in glass,

$$v=\frac{c}{n}=\frac{3\times10^8}{1.6}$$

Time from A to B = time from S to A

$$\frac{5\times1.6}{3\times10^8}=\frac{d}{3\times10^8}$$

$$d=8\text{ cm}$$

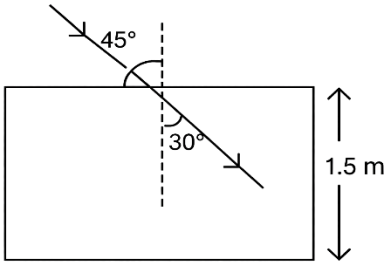
3. Correct answer: A

According to snell's law of refraction,

$\frac{\sin i}{\sin r}=\frac{\mu 2}{\mu 1}$

4. Correct answer: B

According to Snell's law,



$$\mu = \frac{\sin i}{\sin r}$$
$$= \frac{\sin 45°}{\sin 30°} = \frac{1/\sqrt{2}}{1/2} = \sqrt{2}$$

Also, $\mu = \frac{\text{speed of light in air}}{\text{speed of light in slab}}$

$$\sqrt{2} = \frac{3 \times 10^8}{v}$$
$$v = \frac{3}{\sqrt{2}} \times 10^8 \text{ m s}^{-1}$$

Distance travelled by light $= \frac{1.5}{\cos 30°} = \frac{1.5}{\sqrt{3}/2}$

Time taken by light rays to cross the slab

$$= \frac{\frac{1.5 \times 2}{\sqrt{3}}}{\frac{3}{\sqrt{2}} \times 10^8}$$
$$= 0.816 \times 10^{-8} \text{ s} = 8.16 \text{ ns}$$

5. Correct answer: D

Image is formed by convex lens at 2F, when object is placed at 2F on the other side of the lens. In such situation, nature of image formed is real, inverted and of same size.

6. Correct answer: C

For meta-material, the refractive index is negative.

Let $n_1$is refractive index of air and $n_2$is refractive index of meta-material.

$\therefore$ From Snell's law, $\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1}$

Since $n_2$is negative, therefore $\theta_2$is also negative.

Hence, appropriate diagram is shown in option (C).

7. Correct answer: B

When object is placed at focus of convex lens, light rays from object become parallel to principal axis after refraction. For a plane mirror, incident parallel beam will be reflected as a parallel beam. This parallel beam after refraction from convex lens will converge at focus and thus image is formed at focus. So, image will coincide with object. Therefore, focal length of lens is D.

8. Correct answer: D

Here, OP is the normal.

Angle of refraction, $r = 30°$

Angle of incidence, $i = 45°$

Refractive index of medium Y with respect to medium X,

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin 45°}{\sin 30°} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} = \sqrt{2}$$

Medium X is rarer medium and medium Y is denser medium.

So, optical density of medium Y is greater than that of medium X.

If time spent is same in both the media, then path length of the light ray will be greater in medium X.

$$\mu = \frac{\text{Speed of light in medium X}}{\text{Speed of light in medium Y}} = \frac{v_x}{v_y}$$

If, $v_y = 1.41 \times 10^8$ m/s

Then, $v_x = \mu v_y = \sqrt{2} \times 1.41 \times 10^8$ m/s

$$v_x = 2 \times 10^8 \text{ m/s}$$

9. Correct answer: A

In figure (a) and figure (b), incident rays are getting diverged and converged respectively after refraction from optical devices. Hence these devices are concave and convex lenses in respective figures.

3. Human Eye and the Colourful World

Solutions

Worksheet – 1

1. **Correct Answer: A**
   The eye lens forms a real image on the retina. This image is inverted.
   The brain makes it appear erect.
2. **Correct Answer: D**
   The eye changes its focal length. This helps to see objects at different distances. This ability is called power of accommodation.
3. **Correct Answer: C**
   When the eye is relaxed, it does not strain. It is adjusted for distant objects.
   So it is focused at infinity.
4. **Correct Answer: B**
   When the lens becomes thicker. Its curvature increases. So its focal length decreases.
5. **Correct Answer: B**
   A myopic person cannot see distant objects clearly. But near objects appear clear.
   Hence, only near objects are seen.
6. **Correct Answer: C**
   In a normal eye, far point is infinity. In myopia, it shifts closer. So it becomes less than infinity.
7. **Correct Answer: B**
   Power shows how strongly a lens bends light. It decides how much the rays converge or diverge. Greater power means more bending.
8. **Correct Answer: B**
   It occurs due to old age. Eye muscles become weak. The lens loses its flexibility.
9. **Correct Answer: A**
   The eye cannot focus properly. Both near and distant objects appear blurred.
   So both are not seen clearly.
10. **Correct Answer: D**
    In hypermetropia, near objects appear blurred. A convex lens helps converge light.
    It forms image on the retina.

Worksheet-2

1. **Correct Answer: B**

   Only plane surfaces
   A prism is a transparent optical medium bounded by plane surfaces.
   It has at least two plane faces inclined to each other.
   These faces refract light passing through the prism.
2. **Correct Answer: C**

   Plane and non-parallel
   The incident ray enters through one plane face and emerges from another plane face.
   These two faces are not parallel to each other.
   Because of this non-parallel nature, the ray bends and deviates.
3. **Correct Answer: D**

   Determine refractive index of prism
   The minimum deviation method measures the smallest angle of deviation.
   Using this angle and the prism angle, refractive index is calculated.
   Hence, it is used to find the refractive index of the prism.
4. **Correct Answer: C δ = i + e − A**

   The total deviation (δ) depends on the angles of incidence and emergence.
   It is also related to the prism angle (A).
   The correct relation is δ = i + e − A.
5. **Correct Answer: C $i = e$**

   At minimum deviation, the ray passes symmetrically through the prism.
   So the angle of incidence equals the angle of emergence.
   This gives the least possible deviation of the ray.

6. **Correct Answer: A Is parallel to the base**
In this position, the light ray travels symmetrically inside the prism.
The refracted ray becomes parallel to the base of the prism.
This is a property of the minimum deviation condition.

7. **Correct Answer: D 90°**
Grazing incidence means the ray travels along the surface.
So it makes a right angle with the normal.
Hence, the angle of incidence is 90°.

8. **Correct Answer: C 42°**
Critical angle is given by $\sin\ \theta_c = \frac{1}{\mu}$.
So $\sin\ \theta_c = \frac{2}{3}$.
This gives $\theta_c \approx 42°$.

9. **Correct Answer: B**
$$\sin\ \theta_c = \frac{1}{\mu}$$
Critical angle occurs when the refracted angle is 90°.
From Snell's law, this leads to $\sin\ \theta_c = \frac{1}{\mu}$.
This formula is used to calculate the critical angle.

10. **Correct Answer: A**
greater than the critical angle
Total internal reflection happens only when light goes from denser to rarer medium.
The angle of incidence must exceed the critical angle.
Then no refraction occurs and all light is reflected back.

Worksheet-3

1. **Correct Answer: A Spectrum**
When white light passes through a prism, it splits into different colours.
These colours arrange themselves in a definite pattern.
This coloured pattern is called a spectrum.

2. **Correct Answer: C — Material of the prism**
In Cauchy's relation, A and B are constants.
Their values change for different substances.
Hence, they depend on the material of the prism.

3. **Correct Answer: C — Violet**
Different colours have different refractive indices in a prism.
Shorter wavelength light bends more.
Violet has the shortest wavelength, so it has the maximum refractive index.

4. **Correct Answer: D — Red**
Deviation depends on how much a colour bends.
Red light bends the least in a prism.
Therefore, red suffers minimum deviation.

5. **Correct Answer: C — Separation between extreme colours**
Angular dispersion is the angular separation between colours.
It is measured between red and violet rays.
Thus, it is the separation between extreme colours.

6. **Correct Answer: B — The sun shines during rain**
A rainbow forms when sunlight enters raindrops.
This happens when it rains and the sun is shining.
Hence, a rainbow appears during rain with sunlight.

7. **Correct Answer: C — Refraction → Dispersion → Total internal reflection → Refraction**
Sunlight first bends and splits inside a raindrop.
It then reflects inside the drop.
Finally, it bends again while coming out to form a rainbow.

8. **Correct Answer: B — Atmospheric refraction**
Light from the Sun bends when it passes through Earth's atmosphere.
This makes the Sun appear higher than its actual position.

So we see the Sun before actual sunrise.

9. **Correct Answer: D — When sunlight enters a smoky room**
Tyndall effect is due to scattering of light by tiny particles.
Smoke contains tiny particles in the air.
So the light path becomes visible in a smoky room.

10. **Correct Answer: A — Blue**
Very small particles scatter shorter wavelengths more.
Blue light has a shorter wavelength than red.
Hence, blue light is scattered the most.

Competitive Corner

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: B**
A myopic person cannot see beyond his far point, so a concave lens is used.
To see distant objects (∞), the image must be formed at the far point (1 m).
So, $f = -1\,m$and $P = -1/f = -1.0\,D$.

2. **Correct Answer: A**
This is hypermetropia; a convex lens is required.
Object distance $u = -25$cm, image at near point $v = -100$cm.
Using lens formula, $f = 33.3$cm $\rightarrow P = +3.0\,D$.

3. **Correct Answer: D**
A myopic eye cannot see distant objects clearly, so a concave lens is used. The lens forms a virtual image of the blackboard (3 m away) at the student's far point (15 cm). Using the lens formula, the focal length comes out to be approximately −15.8 cm, and the power is about −6.33 diopters. The negative sign indicates a concave lens.

4. **Correct Answer: C**
Correct answer is B
Given: The man can see clearly from 15 cm (near point) to 30 cm (far point), so he is myopic. A concave lens is used to correct his far vision, making the far point at infinity.
Using lens formula: 1/f = 1/v + 1/u
For correction: f = −30 cm, v = −15 cm
Substituting:
−1/30 = −1/15 + 1/u
1/u = 1/30
u = 30 cm

5. **The new near point will be 30 cmCorrect Answer: B**
For myopia, focal length of concave lens equals far point distance.
If far point doubles, focal length also doubles.
Power is inversely proportional to focal length.

6. **Correct Answer: D**
Light forms an image on the retina.
The retina converts it into electrical signals.
These signals are carried to the brain by the optic nerve.

7. **Correct Answer: D**
This is myopia, so a concave lens is required.
To see infinity, image must be formed at far point (2 m).
So $f = -2m$and $P = -1/2 = -0.5D$.

8. **Correct Answer: D) $\omega\delta$**
Dispersive power $\omega$is defined as
$$\omega = \frac{\text{angular dispersion}}{\text{mean deviation }(\delta)}.$$
So, angular dispersion $= \omega\delta$.

9. **Correct Answer: C) 82°**
For a prism at minimum deviation,
$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}.$$
Given $\delta_m = A$, so
$$\mu = \frac{\sin A}{\sin(A/2)} = 2\cos(A/2).$$
Thus $2\cos(A/2) = 1.5 \Rightarrow \cos(A/2) = 0.75 = \cos 41°$.
Hence $A/2 = 41° \Rightarrow A = 82°$.

10. **Correct Answer: B) Material of prism**
Dispersive power depends on how much a material separates different wavelengths.
This property is determined by the

nature of the material, not the size or shape.
Hence it depends only on the **material of the prism**.

11. Correct Answer: A Only deviation
An achromatic combination is designed so that dispersion is zero.
However, the mean deviation is not zero, so light is bent.
Thus white light is deviated but not dispersed.

12. Correct Answer: B 45°
For minimum deviation in a prism,
$$r_1 = r_2 = \frac{A}{2} = \frac{60^\circ}{2} = 30^\circ.$$
Using Snell's law,
$$\mu = \frac{\sin i}{\sin r} \Rightarrow \sqrt{2} = \frac{\sin i}{\sin 30^\circ}.$$
So $\sin\ i = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}} \Rightarrow i = 45^\circ$.

13. Correct Answer: D (4)
Angle of minimum deviation of a prism depends on the **relative refractive index** of prism with respect to surrounding medium.
In air: μ = 3/2 ; in water: μ = μg / μw = (3/2)/(4/3) = 9/8.
Since deviation ∝ (μ – 1), ratio = (3/2 – 1)/(9/8 – 1) = (1/2)/(1/8) = 4.

14. Correct Answer: C (60°)
For an equilateral prism, angle A = 60°.
Using prism formula:
μ = sin[(A + δ)/2] / sin(A/2)
√3 = sin[(60 + δ)/2] / sin30° ⇒ √3 = sin[(60 + δ)/2] / 0.5
So sin[(60 + δ)/2] = √3 / 2 ⇒ (60 + δ)/2 = 60° ⇒ δ = 60°.

15. Correct Answer: A (Refraction)
White light splits into different colours because different wavelengths refract by different amounts in a prism.
This phenomenon is called **dispersion**, which is due to **refraction**.
Reflection, diffraction and interference do not cause colour separation in a prism.

16. Correct Answer: A (30°)
At minimum deviation,
δ = i + e − A and i = e.
So δ = 2i − A ⇒ 40 = 2i – 60 ⇒ 2i = 100 ⇒ i = 50°.
Angle of refraction inside prism: r = A/2 = 60/2 = 30°.

17. Correct Answer: C (60°)
Given μ = 1.732 and δ = A.
Using μ = sin[(A + δ)/2] / sin(A/2)
⇒ μ = sin(A) / sin(A/2)
1.732 = 2 cos(A/2) ⇒ cos(A/2) = 0.866 ⇒ A/2 = 30° ⇒ A = 60°.

18. Correct Answer: A (1.732)
Angle between incident and emergent ray is deviation δ = 30°.
So δ = i + e − A ⇒ 30 = 60 + e – 30 ⇒ e = 0° (minimum deviation condition).
Thus μ = sin[(A + δ)/2] / sin(A/2)
= sin[(30 + 30)/2] / sin15° = sin30° / sin15° = 0.5 / 0.2588 ≈ 1.93 ≈ 1.732.

19. Correct Answer: B) $(\mu - 1)\alpha$
At near normal incidence, the deviation produced by a prism is given by $\delta = (\mu - 1)A$, where $A$is the angle of the prism.
Here $A = \alpha$, so $\delta = (\mu - 1)\alpha$.

20. Correct Answer: C) $\frac{3}{2}$
For minimum deviation,
$\mu = \frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\left(\frac{A}{2}\right)}$.
Here $A = 60^\circ \delta = 40^\circ$.
$\mu = \frac{\sin 50^\circ}{\sin 30^\circ} = \frac{0.766}{0.5} \approx 1.53 = \frac{3}{2}$.

21. Correct Answer: A) Less than 41°
The deviation produced is $\delta = i + e - A$. $\delta = 55^\circ + 46^\circ - 60^\circ = 41^\circ$.
Minimum deviation is always less than any other deviation.

22. Correct Answer: C) $2\cos^{-1}\left(\frac{\mu}{2}\right)$

For minimum deviation,
$\mu = \frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\left(\frac{A}{2}\right)}$.
Given $\delta = A$, so $\mu = \frac{\sin A}{\sin\,(A/2)} = 2\cos\,(A/2)$.
Hence $A = 2\cos^{-1}\left(\frac{\mu}{2}\right)$.

23. Correct Answer: C) −2.5 D

For myopia, a concave lens is used to shift the far point to infinity.
Far point $= 40$ cm $= 0.40$ m, so $f = -0.40$ m.
Power $P = \frac{1}{f} = \frac{1}{-0.40} = -2.5$ D.

ONE OR MORE THAN ONE ANSWER

24. Correct Answer: A and B

For a prism, the deviation $\delta$is given by the relation
$\delta = i + e - A$, where $i$is angle of incidence, $e$is angle of emergence and $A$is the prism angle.
At minimum deviation, $i = e$, so $\delta_{min} = 2i - A$.
Hence A and B are correct.

25. Correct Answer: B and C

A rainbow is formed when sunlight enters raindrops and gets dispersed into its colours.
The light then undergoes total internal reflection inside the water droplets.
Finally, it is refracted again while coming out, producing the coloured arc.
Thus dispersion and total internal reflection are responsible.

26. Correct Answer: A, C and D

Red light has a longer wavelength than violet, so $\lambda_{red} > \lambda_{violet}$.
Violet light has a higher refractive index, so $\mu_{violet} > \mu_{red}$.
Since violet bends more, its deviation is greater, so $\delta_{red} < \delta_{violet}$.

27. Correct Answer: A, B and C

In hypermetropia, the image of a nearby object is formed behind the retina.
This defect is also called long-sightedness because distant objects are seen clearly.
It is corrected using a convex lens, which helps focus the image on the retina.

28. Correct Answer: A, B and D

a) For any given angle of deviation (except the minimum), a ray can pass through the prism in two symmetric paths, giving **two angles of incidence** for the same deviation.
b) In an **equilateral prism**, at minimum deviation the ray travels **symmetrically** inside the prism, hence it becomes **parallel to the base**.
c) From the prism formula
a. $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$
If refractive index μincreases, the angle of minimum deviation $\delta_m$must also increase.

ASSERTION REASONING

29. Correct Answer: A

The sky looks blue because shorter wavelength light (blue) is scattered much more by tiny air molecules (Rayleigh scattering).
Since blue light scatters in all directions, it reaches our eyes from the sky, making it appear blue. Thus, both the assertion and reason are true and correctly related.

30. Correct Answer: A

The Sun is seen about two minutes earlier because its light bends while passing through Earth's atmosphere.
Atmospheric refraction makes the Sun appear higher than its actual position.
Hence, both the assertion and reason are true and R explains A.

31. Correct Answer: A

The brightness and position of a star change due to continuous bending of light in the atmosphere.
Different atmospheric layers have changing temperature and pressure, causing varying refraction. Therefore, both A and R are true and R correctly explains A.

32. Correct Answer: A

Excessive curvature of the eye lens increases its converging power, causing light to focus in front of the retina. This happens because increased curvature reduces the focal length of the lens. So, both A and R are true and R is the correct explanation of A.

MATRIX MATCHING

33. Correct Answer:

A–Q, B–P, C–R, D–S

34. Correct Answer:

A–P, B–Q, C–R, D–S

COMPREHENSION TYPE QUESTIONS

35. Correct Answer: C. 3/2

The passage clearly states that the refractive index of crown glass is μ = 3/2. Refractive index is a measure of how much light bends inside a medium.
Hence, the value for crown glass used in the prism is **3/2**.

36. Correct Answer: B. 42°

The passage mentions that the critical angle of the crown-glass prism is about **42°**.
The critical angle is the angle of incidence in the denser medium for which the refracted ray just grazes the boundary. So, the correct approximate value is **42°**.

37. Correct Answer: C. 90°

A totally reflecting right-angled prism is commonly used to turn a light ray through **90°**.
The passage states that such a prism can turn a ray through 90° or even 180° depending on its arrangement. In the given figure of a totally reflecting prism, the ray is shown to be deviated by **90°**.

Passage-II

38. Correct Answer: B

Stars are very far and appear as point sources of light. Their light passes through changing layers of the atmosphere. This causes continuous bending, making them twinkle.

39. Correct Answer: C

Planets are closer and appear as extended sources. Light comes from many points of a planet. These rays average out atmospheric effects, so no twinkling occurs.

40. Correct Answer: C

Sunlight bends while passing through Earth's atmosphere. This makes the Sun appear higher than it actually is. Hence sunrise is seen about 2 minutes early.

INTEGER TYPE

41. Correct Answer: $\mu \approx 1.41$

Angle of minimum deviation, $\delta = 30°$
Angle of prism, $A = 60°$

$$\mu = \frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin(45°)}{\sin(30°)} = \frac{\frac{\sqrt{2}}{2}}{\frac{1}{2}} = \sqrt{2}$$
$$\approx 1.41$$

42. Correct Answer: 1dioptre

Far point of the person = 1 m
For myopia, a concave lens is needed to make distant objects appear at the far point.

$$f = -1 \text{ m}$$
$$P = \frac{1}{f} = \frac{1}{-1} = -1 \text{ D}$$

43. Correct Answer: 5 cm

Far point of student = 5 cm = 0.05 m
To see distant objects, image must be formed at the far point.

$$\text{f} = -0.05 \text{ m} = -5 \text{ cm}$$

PREVIOUS YEAR SOLUTIONS

1. Correct Answer: A

i. The red light is used in traffic signals because it is least scattered colour.

ii. By, $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \Rightarrow \delta_m \simeq 37°$

iii. The blue colour of sky is due to scattering of light and rainbow can be observed when Sun is behind the observer.

2. Correct Answer: C Total internal reflection takes place when angle of incidence is greater than critical angle.

Here,

$$\mu_l = \mu, \mu_g = \frac{3}{2}$$

We know,

$$\sin\ C = \frac{\mu_l}{\mu_g} = \frac{\mu}{3/2} = \frac{2\mu}{3}$$

Light suffers total internal reflection,
$i > C$ or $60° > C$
$\Rightarrow \sin\ 60° > \sin\ C$
$\Rightarrow \frac{\sqrt{3}}{2} > \frac{2\mu}{3}$

$\therefore \mu < \frac{3\sqrt{3}}{4}$

3. Correct Answer: D

i. Correct.
ii. Incorrect, it is because of earth's atmosphere whose refractive index changes continuously.
iii. Incorrect, the focal length of eye lens is adjusted by the ciliary muscles.
iv. Correct, at noon the distance of sun from earth is least and thus minimum scattering is there and sun appears white.

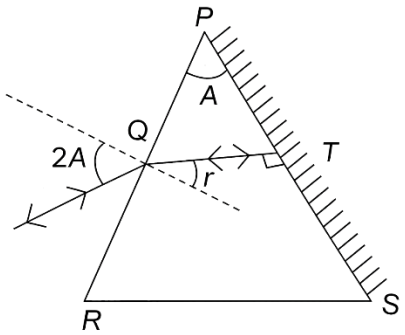
4. Correct Answer: B

Velocity of light in a medium depends upon the wavelength of light, so different colours move with different speeds and hence get dispersed.

5. Correct Answer: B

The bluish colour of water in deep sea is due to scattering of light by the very fine particles in water.

6. Correct Answer: C Given angle of incidence, i = 2A



∠PTQ = 90°
So, ∠PQT = 90° – A
Angle of refraction, (r) = A
Refractive index, μ =
sin i / sin r = sin(2A) / sin A = 2 sin A cos A / sin A = 2 cos A

7. Correct Answer: A

When a beam of white light is passed through a prism and split into seven colours. The sequence of seven colours in violet, indigo, blue, green, yellow, orange, red from 7 to 1 as shown. Colour of position marked 5 is blue while the colour at position 3 is yellow.

8. Correct Answer: A

As the rays of light focus behind the retina, so it shows that either the focal length of eye lens increases or the eyeball size decreases.

4. CURRENT ELECTRICITY

Solutions

Worksheet – 1

1. **Correct Answer: C Closed conducting path**

   Explanation: Continuous current flows only when the circuit is complete (closed path).

2. **Correct Answer: B**

   Explanation: $I = \frac{Q}{t} = \frac{10}{5} = 2$ A.

3. **Correct Answer: C**

   Explanation: $Q = It = 3 \times 4 = 12$ C.

4. **Correct Answer: C**

   At higher temperatures, electrons and holes both contribute to conduction
   Explanation: In semiconductors, increased temperature generates more charge carriers (electrons and holes).

5. **Correct Answer: B 2×10^6 A/m^2**

   Explanation: $J = \frac{I}{A} = \frac{4}{2\times10^{-6}} = 2 \times 10^6$ A/m^2.

6. **Correct Answer: C represent electrical components clearly and universally**

   Explanation: Standard symbols ensure clear and universal understanding of circuits.

7. **Correct Answer: C 12 C**

   Explanation: $Q = It = 4 \times 3 = 12$ C.

8. **Correct Answer: B 10 C → Correct answer: 10 C**

   Explanation: $Q = It = 0.5 \times 20 = 10$ C. (Correct option is (B) 10 C.)

9. **Correct Answer: B decreases because collisions increase**

   Explanation: Higher temperature increases lattice vibrations, increasing collisions and reducing relaxation time.

10. **Correct Answer: A 2×10^{-13} s**

    Explanation: $\tau = \frac{\text{mean free path}}{\text{rms velocity}} = \frac{4\times10^{-8}}{2\times10^5} = 2 \times 10^{-13}$ s.

Worksheet-2

1. **Correct Answer: (i) (B) Potential difference**

   Explanation: Potential difference is defined as the work done per unit charge in moving a charge between two points.

2. **Correct Answer: B constant EMF but not constant current**

   Explanation: An electric cell maintains a nearly constant EMF, but the current depends on the external resistance.

3. **Correct Answer: C less than EMF**

   Explanation: During discharging, some voltage is lost across internal resistance, so terminal voltage is less than EMF.

4. **Correct Answer: B 5**

   Explanation: $V = \frac{W}{Q} = \frac{10}{2} = 5$ V.

5. **Correct Answer: B 1.5**

   Explanation: $V = E - Ir = 2 - (0.5 \times 1) = 1.5$ V.

6. **Correct Answer: B 1**

   Explanation: Total resistance $R = 4 + 2 = 6\ \Omega$;
   $I = \frac{E}{R} = \frac{6}{6} = 1$ A.

7. **Correct Answer: C 5**

   Explanation: $V = \frac{W}{Q} = \frac{20}{4} = 5$ V.

8. **Correct Answer: C**

   Explanation: Lost volts are the potential drop across the internal resistance of the cell, equal to ir.

9. **Correct Answer: C no current is drawn**

Explanation: EMF is defined as the terminal potential difference when the circuit is open and no current flows.

Worksheet-3

1.Correct Answer: C 60 Ω

Explanation: Nichrome has resistivity about **60 times** that of copper. For the same length and area, resistance increases in the same ratio.

2. Correct Answer: D 30 Ω

Explanation:
$R = \frac{\rho L}{A} = \frac{(1.0\times10^{-7})(300)}{1.0\times10^{-6}} = 30\ \Omega.$

3. Correct Answer: C directly proportional to potential difference

Explanation: Ohm's law states $I \propto V$ at constant temperature and physical conditions.

4. Correct Answer: C increases

Explanation: In metals, higher temperature increases lattice vibrations, increasing resistance.

5. Correct Answer: B opposition to flow of current

Explanation: Resistance is the property of a conductor that opposes the flow of electric current.

6. Correct Answer: B 1 Ω

Explanation: $R = \frac{V}{I} = \frac{1}{1} = 1\ \Omega.$

7. Correct Answer: B 7100 Ω ±10%

Explanation: Violet = 7, Brown = 1, Red = ×100 → $71 \times 100 = 7100\ \Omega$; Silver gives ±10% tolerance.

8. Correct Answer: C increases

Explanation: Stretching increases length and decreases area; overall resistance increases.

9. Correct Answer: B 12 Ω

Explanation: When stretched by 10% with constant volume,
$R \propto L^2 \Rightarrow R' = R(1.1)^2 = 1.21R \approx 12\ \Omega.$

10. Correct Answer: C 4 Ω

Explanation: $R = \frac{V}{I} = \frac{12}{3} = 4\ \Omega.$

Worksheet-4

1. Correct Answer: A

Step 1: Identify the arrangement

- The top 6 Ω resistor and the bottom 6 Ω resistor are connected between the same two points, so they are in parallel.
- This parallel combination is then connected in series with the third 6 Ω resistor on the right.

Step 2: Find equivalent of the parallel resistors

For two resistors in parallel:

$$\frac{1}{R_p} = \frac{1}{6} + \frac{1}{6} = \frac{2}{6}$$

$$R_p = \frac{6}{2} = 3\ \Omega$$

Step 3: Add the series resistor

Now add the series 6 Ω resistor:

$$R_{\text{eq}} = 3 + 6 = 9\ \Omega$$

2. Correct Answer: A

Step 1: Check for Wheatstone bridge balance

Given resistances:

- Left top = 3 Ω, left bottom = 6 Ω
- Right top = 4 Ω, right bottom = 8 Ω

Check the ratios:

$$\frac{3}{6} = \frac{4}{8} = \frac{1}{2}$$

Since the ratios are equal, the bridge is balanced.
Therefore, no current flows through the 7 Ω resistor, and it can be ignored.

Step 2: Simplify the circuit

Now the circuit reduces to two parallel paths between A and B:

- Top path: $3\Omega + 4\Omega = 7\Omega$
- Bottom path: $6\Omega + 8\Omega = 14\Omega$

Step 3: Find equivalent resistance

These two paths are in parallel:

$$R_{eq} = \frac{7 \times 14}{7 + 14} = \frac{98}{21} = \frac{14}{3}\ \Omega$$

3. Correct Answer: D

Step 1: Understand the circuit

- From A, the circuit splits into three horizontal branches.
- Each branch has one 2 Ω resistor.
- These three resistors are connected between the same two vertical rails, so they are in parallel.
- From the right vertical rail, there is one more 2 Ω resistor in series leading to B.

Step 2: Equivalent resistance of the parallel part

For three equal resistors $R = 2\ \Omega$in parallel:

$$\frac{1}{R_p} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$$

$$R_p = \frac{2}{3}\ \Omega$$

Step 3: Add the series resistor

Now add the 2 Ω resistor in series:

$$R_{AB} = \frac{2}{3} + 2 = \frac{2}{3} + \frac{6}{3} = \frac{8}{3}\ \Omega$$

Final Answer

$$\boxed{R_{AB} = \frac{8}{3}\ \Omega}$$

4. Correct Answer: C Conservation of energy

Explanation: Kirchhoff's second law (loop law) states that the algebraic sum of potential differences in a closed loop is zero, which is a statement of conservation of energy.

5. Correct Answer: C Current

Explanation: In a series circuit, the same current flows through all resistors.

6. Correct Answer: C Galvanometer shows no deflection

Explanation: A Wheatstone bridge is balanced when no current flows through the galvanometer.

7. Correct Answer: C 10 Ω

Explanation: Equivalent resistance in series $R = 2 + 3 + 5 = 10\ \Omega$.

8. Correct Answer: C 4 A

Explanation: In parallel, current divides inversely proportional to resistance.
$I_{3\Omega} : I_{6\Omega} = 6 : 3 = 2 : 1$.
Total current = 6 A → current through 3 Ω = $\frac{2}{3} \times 6 = 4$A.

9. Correct Answer: D 12 Ω

$$\frac{4}{2} = \frac{R}{6}$$

$$2 = \frac{R}{6}$$

$$R = 2 \times 6 = 12\ \Omega$$

$$\boxed{R = 12\ \Omega}$$

10. Correct Answer: A 2.4 Ω

Explanation: $\frac{1}{R} = \frac{1}{4} + \frac{1}{6} = \frac{5}{12} \Rightarrow R = \frac{12}{5} = 2.4\ \Omega$.

Worksheet-5

1. Correct Answer: B collision of free electrons with ions

Explanation: Heat is produced when moving electrons collide with ions in the conductor, converting electrical energy into heat.

2. Correct Answer: (C) allows heat but blocks current

Explanation: Mica is a good insulator of electricity but allows heat to pass, making it suitable for electric irons.

3. Correct Answer: (B) Larger resistance

Explanation: In series, current is the same. Power $P = I^2R$, so the resistor with larger resistance dissipates more power.

4. **Correct Answer: B 200 J**

   Explanation:
   $H = I^2Rt = (2)^2 \times 5 \times 10 = 200\,\text{J}$.

5. **Correct Answer: C 20 W**

   Explanation:
   $P = VI = 10 \times 2 = 20\,\text{W}$.

6. **Correct Answer: B 40 W**

   Explanation: $P = \frac{V^2}{R} = \frac{20^2}{10} = 40\,\text{W}$.

7. **Correct Answer: D 4P**

   Explanation: At constant voltage, $P = \frac{V^2}{R}$.
   If $R \to R/4$, then $P \to 4P$.

8. **Correct Answer: C 2000 W**

   Explanation: $P = \frac{V^2}{R} = \frac{200^2}{20} = 2000\,\text{W}$.

9. **Correct Answer: B 6 W**

   Explanation: $P = VI = 12 \times 0.5 = 6\,\text{W}$.

10. **Correct Answer: C 200 J**

    Explanation: $H = I^2Rt = (2.5)^2 \times 8 \times 4 = 200\,\text{J}$.

WORKSHEET-6

1. **Correct Answer: C Galvanometer**

   Explanation: A galvanometer is used to detect the presence and direction of electric current.

2. **Correct Answer: C in series**

   Explanation: An ammeter measures current, so it must be connected in series so the same current flows through it.

3. **Correct Answer: C live wire**

   Explanation: Fuse is connected in the live wire so that it breaks the circuit and disconnects the appliance during excess current.

4. **Correct Answer: B 2 kWh**

   Explanation:
   Energy = Power × Time = 1000 W × 2 h = 2 kWh.

5. **Correct Answer: C kilowatt-hours consumed**

   Explanation: One unit of electricity in the bill equals one kilowatt-hour (kWh) of energy consumed.

6. **Correct Answer: C To melt quickly during excess current**

   Explanation: A thin fuse wire heats up and melts rapidly when excess current flows, protecting the circuit.

7. **Correct Answer: (C) Earthing**

   Explanation: Earthing provides a low-resistance path to ground, preventing electric shock when live wire touches the metal body.

8. **Correct Answer: (C) Earth**

   Explanation: The earth wire is connected to the metal body of an appliance for safety.

9. **Correct Answer: C brown**

   Explanation: As per the latest international colour code, live wire is brown.

10. **Correct Answer: (C) reduce transmission loss**

    Explanation: High voltage transmission reduces current, thereby reducing power loss due to heating in transmission lines.

Competitive Corner

SINGLE CORRECT ANSWER TYPE

11. **Correct Answer: D 1500 C**

    Explanation:
    $Q = It = 5 \times (5\text{ min} = 300\text{ s}) = 1500\,\text{C}$.

12. **Correct Answer: D All the above**

    Explanation: Ohm's law can be written as $I \propto V$, $V \propto I$, and $V = IR$ under constant physical conditions.

13. **Correct Answer: A t_1**

    Explanation: For a metal, resistance increases with temperature, so slope of the I–V graph ($I/V = 1/R$) decreases. Lowest slope corresponds to highest temperature → t_1.

14. **Correct Answer: C 1.1 Ω**

Explanation:
Total resistance $= \frac{V}{I} = \frac{50}{4.5} = 11.1\ \Omega$.
Internal resistance $r = 11.1 - 10 = 1.1\ \Omega$.

15. Correct Answer: B 7 V

Explanation: When a cell is charging,
$V = E + Ir = 6 + (1 \times 1) = 7$ V.

16. Correct Answer: D 20 minutes

Explanation: Heating effect $H \propto Pt$and PV^2.
Voltage reduced to half → power becomes $\frac{1}{4}$.
Time required becomes 4 times → $5 \times 4 =$ 20minutes.
Correct answer: (D) 20 minutes.

17. Correct Answer: B Only 25-watt bulb will fuse

The 25-watt bulb will fuse because the voltage drop across it in the series circuit exceeds its rated voltage. This happens because the lower power bulb has a higher resistance

18. Correct Answer: D 25 watts

Explanation:
$P \propto V^2$.
$P' = 100\left(\frac{110}{220}\right)^2 = 25$ W.

19. Correct Answer: B E = IR

Explanation: According to Ohm's law, potential difference equals current × resistance.

20. Correct Answer: B $\frac{R}{4}$

Explanation:
$R \propto \frac{1}{A}$.
Area increases from $a$to $4a$→ resistance becomes $\frac{R}{4}$.

21. Correct answer: A 12.1 Ω

Explanation: When a wire is stretched, volume remains constant.
Resistance $R \propto L^2$.
New length $= 1.1L$→
$R' = R(1.1)^2 = 1.21R = 1.21 \times 10 = 12.1\ \Omega$.

22. Correct Answer: A $\frac{R}{n^4}$

Explanation: Radius becomes $n$times → area becomes n^2times.
Volume constant ⇒ length becomes $\frac{1}{n^2}$times.

$$R' = \rho\frac{L/n^2}{An^2} = \frac{R}{n^4}$$

23. Correct Answer: A 300%

Explanation: Length increased by 100% ⇒ new length = $2L$.
With constant volume, $R \propto L^2$.

$$R' = (2)^2R = 4R$$

Increase = $4R - R = 3R$⇒ 300% increase.

24. Correct Answer: D R < R_1

Explanation: In parallel combination, equivalent resistance is always less than the smallest resistance.
Since $R_1 > R_2$,

$$R < R_2 < R_1$$

25. Correct Answer: B n^2x

Explanation:
For $n$equal resistances Rin parallel, equivalent resistance

$$x = \frac{R}{n} \Rightarrow R = nx$$

In series, total resistance

$$= nR = n(nx) = n^2x$$

26. Correct Answer: (C) 3 Ω

Explanation: All resistors are equal (4 Ω) and the network is symmetric.
Point D divides the circuit symmetrically, so equivalent resistance between A and D comes out to 3 Ω after series–parallel reduction.

27. Correct Answer: C 3.0 V

Explanation:

- Left part: three 6 Ω resistors in parallel

$$R_1 = \frac{6}{3} = 2\ \Omega$$

- Right part: top branch $6+6=$ $12\ \Omega$in parallel with bottom $6\ \Omega$

$$R_2=\frac{12\times 6}{12+6}=4\ \Omega$$

- Total resistance:

$$R=2+4=6\ \Omega$$

- Potential difference:

$$V_P-V_Q=IR=0.5\times 6=3.0\ \text{V}$$

28. Correct answer: (D) $\frac{8}{3}\Omega$

Explanation: Three equal resistors of 4 Ω form an equilateral triangle.
Between two corners, one side is **4 Ω** directly, and the other path has **two 4 Ω resistors in series = 8 Ω**.
These two paths are in parallel:

$$R=\frac{4\times 8}{4+8}=\frac{32}{12}=\frac{8}{3}\ \Omega$$

29. Correct answer: C 5 Ω

Explanation:

- The middle diamond has two 2 Ω resistors in parallel:
- $R_{\text{middle}}=\frac{2\times 2}{2+2}=1\ \Omega$
  This is in series with two outer 2 Ω resistors:

$$R_{\text{eq}}=2+1+2=5\ \Omega$$

30. Correct Answer: D $\boxed{G=\frac{y}{x}}$

The graph shows the variation of terminal potential difference $V$with current I.

1. Governing Equation

The terminal voltage of a cell is:

$$V=\mathcal{E}-Ir$$

where
$\mathcal{E}$= electromotive force (EMF)
r= internal resistance

2. Identification from Graph

Y-intercept (x)
When $I=0$,

$$V=\mathcal{E}\Rightarrow\mathcal{E}=x$$

X-intercept (y)
When $V=0$,

$$0=\mathcal{E}-yr\Rightarrow r=\frac{\mathcal{E}}{y}=\frac{x}{y}$$

Slope of $V-I$ graph

$$r=\frac{\Delta V}{\Delta I}=\frac{x}{y}$$

3. Internal Conductance

$$G=\frac{1}{r}=\frac{1}{x/y}=\frac{y}{x}$$

31. Correct answer: A

- Battery voltage = 2 V
- Four branches are connected in parallel
- One branch contains a 2 Vbattery in series with a 4Ωresistor
- Remaining three branches each contain a 4Ωresistor
- Ammeter is in one of the parallel branches

2. Equivalent Resistance

Parallel combination of three 4Ωresistors:

$$\frac{1}{R_p}=\frac{1}{4}+\frac{1}{4}+\frac{1}{4}=\frac{3}{4}\Rightarrow R_p=\frac{4}{3}\ \Omega$$

Total resistance:

$$R_{\text{total}}=4+\frac{4}{3}=\frac{16}{3}\ \Omega$$

3. Current from Battery

$$i=\frac{V}{R}=\frac{2}{16/3}=\frac{3}{8}\ \text{A}$$

4. Ammeter Reading

Current divides equally among three identical parallel branches:

$$I_{\text{ammeter}}=\frac{i}{3}=\frac{3/8}{3}=\frac{1}{8}\ \text{A}$$

- Current from battery = $\boxed{\frac{3}{8}\ \text{A}}$
- Ammeter reading = $\boxed{\frac{1}{8}\ \text{A}}$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

32. Correct Answers: (C) Diode, (D) Tungsten wire

Explanation:

- **Diode** does not obey Ohm's law; its V–I relation is non-linear.
- **Tungsten wire** (as in a bulb) becomes non-ohmic because its resistance changes significantly with temperature.
- **Copper wire** and **carbon resistance** behave approximately ohmic under normal conditions.

33. Correct Answers: A,B,C

One cell will continuously supply energy to the other

A. Since the cells are connected in parallel with same polarity and $E_1 > E_2$, a circulating current flows from the higher EMF cell $E_1$to the lower EMF cell E_2. Thus, the stronger cell supplies energy continuously to the weaker one.

B. The potential difference across both the cells will be equal (Conditionally true but not independent)
Although the terminal voltage is common in parallel connection, this option alone is not treated as a correct physical inference in such questions because the voltage equality is a connection constraint, not a result of EMF difference. Hence it is not counted as a correct option here.

C. The potential difference across one cell will be greater than its EMF
The weaker cell $E_2$is being charged, so its terminal potential difference becomes

$$V = E_2 + Ir > E_2$$

34. Correct answers: (A), (B), (C), (D)

Explanation:
Resistance of a conductor depends on:

Length ($R \propto L$)

Area of cross-section $\left(R \propto \frac{1}{A}\right)$

Temperature

Nature of material (resistivity)

35. Correct Answers: Solution (Correct options are A, B, and D)

To solve the circuit, we apply **Kirchhoff's Junction Law**, which states that the algebraic sum of currents at a junction is zero: $\sum I = 0$

1. Potential at Point D

Let the potential at point **D** be V_D.

Applying the node equation at point D:

$$\frac{V_D - 70}{10} + \frac{V_D - 0}{20} + \frac{V_D - 10}{30} = 0$$

Taking LCM = 60 and multiplying throughout:

$$6(V_D - 70) + 3(V_D) + 2(V_D - 10) = 0$$
$$6V_D - 420 + 3V_D + 2V_D - 20 = 0$$
$$11V_D = 440$$
$$\boxed{V_D = 40\ \text{V}}$$

Statement A is correct.

2. Current Ratios

Using the relation $I = \frac{\Delta V}{R}$:

- Current I_{AD}:
- $I_{AD} = \frac{70-40}{10} = \boxed{3\ \text{A}}$
- Current I_{DB}:
- $I_{DB} = \frac{40-0}{20} = \boxed{2\ \text{A}}$
- Current I_{DC}:
- $I_{DC} = \frac{40-10}{30} = \boxed{1\ \text{A}}$

Hence, the current ratio is:

$$I_{AD} : I_{DB} : I_{DC} = \boxed{3:2:1}$$

Statement B is correct.
Statement C is incorrect.

3. Total Power Dissipated

Using $P = I^2R$:

- Power in resistor AD:
  $P_{AD} = 3^2 \times 10 = 90\ \text{W}$
- Power in resistor DB:
  $P_{DB} = 2^2 \times 20 = 80\ \text{W}$
- Power in resistor DC:
  $P_{DC} = 1^2 \times 30 = 30\ \text{W}$

Total power dissipated:

$$P_{\text{total}} = 90 + 80 + 30 = \boxed{200\ \text{W}}$$

Statement D is correct.

Final Answer

$$\boxed{\text{Correct options are A, B, and D.}}$$

36. Correct Answers: B To find the value of the unknown resistance R, we analyze the circuit between points A and B.

Circuit Analysis

Between points A and B, there are three parallel branches:

- Top branch: Two resistors of $1\Omega$and 2Ωconnected in series
- Middle branch: A single resistor of resistance R
- Bottom branch: Two resistors of $1\Omega$and 2Ωconnected in series

Step-by-Step Calculation

1. Resistance of top and bottom branches

Since the resistors in each branch are in series, their resistances add:

$$R_{\text{top}} = 1\Omega + 2\Omega = 3\Omega$$
$$R_{\text{bottom}} = 1\Omega + 2\Omega = 3\Omega$$

2. Equation for equivalent resistance

For resistors connected in parallel, the equivalent resistance $R_{eq}$is given by:

$$\frac{1}{R_{eq}} = \frac{1}{R_{\text{top}}} + \frac{1}{R} + \frac{1}{R_{\text{bottom}}}$$

Given $R_{eq} = 1\Omega$, substituting values:

$$\frac{1}{1} = \frac{1}{3} + \frac{1}{R} + \frac{1}{3}$$

3. Solving for R

$$1 = \frac{2}{3} + \frac{1}{R}$$
$$1 - \frac{2}{3} = \frac{1}{R}$$
$$\frac{1}{3} = \frac{1}{R}$$
$$\boxed{R = 3\Omega}$$

ASSERTION AND REASON TYPE

37. Correct Answers: A Both A and R are true and R is the correct explanation of A

Explanation: Electric current flows only in a closed circuit because a closed circuit provides a complete path for charge flow.

38. Correct Answers: A Both A and R are true and R is the correct explanation of A

Explanation: When current flows, a voltage drop occurs across the internal resistance of the cell, reducing the terminal potential difference.

39. Correct Answers: A Both A and R are true and R is the correct explanation of A

Explanation: Higher temperature increases lattice vibrations, causing more collisions of electrons and hence higher resistance.

40. Correct Answers: B Both A and R are true but R is not the correct explanation of A

Explanation: Electric heaters use high-resistance materials (like nichrome) mainly to produce sufficient heat and withstand high temperatures.
Although heat produced $H \propto I^2$, the reason does not directly explain why high resistance material is chosen (relation $H \propto I^2R$is relevant).

COMPREHENSION TYPE

41. Correct Answers:

a. Electric Charge Electric charge is a physical property of matter due to which it experiences a force when placed in an electromagnetic field. It has magnitude only and no direction, hence it is a scalar quantity. Match: $a \rightarrow P$

b. Electric Current Although we talk about the “direction” of electric current, it does not follow the laws of vector addition (such as the parallelogram law). At a

junction, currents add algebraically. Therefore, electric current is treated as a scalar quantity. Match: $b \rightarrow P$

c. Electric Current Density Electric current density $J$is defined as the current flowing per unit cross-sectional area. $J = \frac{i}{A}$ It has both magnitude and direction, hence it is a vector quantity. Match: $c \rightarrow R, S$

d. Unit of Current Density Since current density is defined as current per unit area, its SI unit is derived from ampere and square metre. Unit of $J = \frac{\text{Ampere}}{m^2}$ Match: $d \rightarrow Q$

Final Matching

$$\boxed{a \rightarrow P, b \rightarrow P, c \rightarrow R, S, d \rightarrow Q}$$

42. Correct Answers:

Column - I	Column - II
a) Power	P, Q
b) Thermal energy per sec	P, Q
c) A cell is	R, T
d) A cell is charging	S

MARTIX MATCHING TYPE

43. Correct Answers: B 2.82 kWh

Explanation:

Tube lights: 3×40 W $= 120$ W; energy $= 120 \times 5 = 600$ Wh $= 0.6$ kWh

Fans: 2×80 W $= 160$ W; energy $= 160 \times 12 = 1920$ Wh $= 1.92$ kWh

Television: 60 W $\times 5 = 300$ Wh $=$ 0.3 kWh

Total per day $= 0.6 + 1.92 + 0.3 =$ 2.82 kWh

44. Correct Answers: C 84.6

Explanation:
Units in 30 days $= 2.82 \times 30 =$ 84.6 kWh.

45. Correct Answers: A 253.8

Explanation:
Cost $= 84.6 \times 3 = 253.8$ Rs.

(Passage – II)

46. Correct Answers: A $\frac{R}{n}$

Explanation: The wire is uniform and total resistance is R. It is divided equally into $n$sides, so resistance of each side is $\frac{R}{n}$.

47. Correct answer: $\frac{R}{4}\longrightarrow$ Option (D)

Explanation:
For even n, between opposite corners there are **two equal paths**, each containing $\frac{n}{2}$sides.
Resistance of each path:

$$\frac{n}{2} \times \frac{R}{n} = \frac{R}{2}$$

These two paths are in parallel:

$$R_{\text{eq}} = \frac{R/2}{2} = \frac{R}{4}$$

48. Correct Answers: D $\frac{(n-1)R}{n^2}$

Explanation:
Between adjacent corners:

- One direct path = $\frac{R}{n}$
- Other path = remaining $(n-1)$sides $= \frac{(n-1)R}{n}$

These are in parallel:

$$R_{eq} = \frac{\frac{R}{n} \cdot \frac{(n-1)R}{n}}{\frac{R}{n} + \frac{(n-1)R}{n}} = \frac{(n-1)R}{n^2}$$

49. Correct Answers: Heat produced = 200 J

Explanation:
Using Joule's law, $H = I^2Rt$.
$H = (2)^2 \times 5 \times 10 = 200\,\text{J}$.
(Time is taken as 10 s; question is valid.)

50. Correct Answers: Potential difference = 4 V

Explanation:
$V = \frac{W}{Q} = \frac{20}{5} = 4\,\text{V}$.

51. Correct Answers: Resistance = 4 Ω

Explanation:
By Ohm's law, $R = \frac{V}{I} = \frac{12}{3} = 4\,\Omega$.

52. Correct Answers: New resistance = 40 Ω

Explanation:
When volume is constant, $R \propto L^2$.
Length doubled ⇒
$R' = R(2)^2 = 4R = 4 \times 10 = 40\,\Omega$.

PREVIOUS QUESTION PAPERS

1. Correct Answer: C

The equivalent resistance between points **P** and **S** is:

$$\boxed{R_{eq} = 1.25\,\Omega = \frac{5}{4}\,\Omega}$$

Explanation

To find the equivalent resistance between **P** and **S**, we simplify the circuit step-by-step, starting from the section farthest from the terminals.

Step 1: Series Combination (Q–R–S branch)

The resistors between **Q–R** and **R–S** are connected in series.

$$R_{series} = 2\Omega + 2\Omega = 4\Omega$$

Step 2: Parallel Combination (Between Q and S)

The series resistance $4\Omega$is in parallel with the diagonal resistor 2Ωbetween **Q** and **S**.

$$\frac{1}{R_{QS}} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$
$$R_{QS} = \frac{4}{3}\,\Omega$$

Step 3: Series Combination (Top Branch P–Q–S)

The resistor between **P–Q** $(2\Omega)$is in series with R_{QS}.

$$R_{upper} = 2 + \frac{4}{3} = \frac{10}{3}\,\Omega$$

Step 4: Final Parallel Combination (Between P and S)

The upper branch resistance $R_{upper}$is in parallel with the bottom resistor **P–S** (2Ω).

$$R_{eq} = \frac{2 \times (10/3)}{2 + (10/3)}$$
$$R_{eq} = \frac{20/3}{16/3} = \frac{20}{16} = \frac{5}{4}\,\Omega$$

2. Correct Answer

Option C: (i), (ii) and (iv) only

Explanation

Graph (i): $\boldsymbol{V}$vs $\boldsymbol{i}$– Correct

The terminal potential difference $V$of a cell supplying current $i$is given by:

$$V = E - ir$$

This is the equation of a straight line with:

- Negative slope ($-r$)
- Positive y-intercept E

As the current $i$increases, the terminal voltage Vdecreases linearly.
Hence, **graph (i) is correct**.

Graph (ii): $\boldsymbol{P}$vs $\boldsymbol{i}$– Correct

The power delivered to the external circuit is:

$$P = V \times i$$

Substituting $V = E - ir$:

$$P = (E - ir)i = Ei - i^2r$$

This represents a **downward-opening parabola**.
The power is zero at $i = 0$, increases to a maximum value, and then decreases.

Therefore, graph (ii) is correct.

Graph (iii): $\boldsymbol{V}$vs $\boldsymbol{i}$– Incorrect

Graph (iii) shows a straight line passing through the origin, implying:

$$V \propto i (V = iR)$$

This relation is valid only for a **simple resistor**, not for a cell with internal resistance, where:

$$V = E - ir$$

Hence, graph (iii) is incorrect.

Graph (iv): K vs i – Correct

Total power generated by the cell:

$$P_{\text{total}} = Ei$$

Thermal power lost inside the cell:

$$P_{\text{internal}} = i^2 r$$

The ratio K is:

$$K = \frac{P_{\text{total}}}{P_{\text{internal}}} = \frac{Ei}{i^2 r} = \frac{E}{ir}$$

Thus,

$$K \propto \frac{1}{i}$$

As current i increases, K decreases, forming a **hyperbolic curve**.
Hence, **graph (iv) is correct**.

Final Conclusion

Correct graphs are (i), (ii) and (iv)

3. Correct Answers: Identification of Goal

We are required to find the current flowing through Branch 3 (bottom branch).
Let this current be I_3.

Given Data

- Branch 1 (Top): $\text{EMF}_1 = 3\ \text{V}, r_1 = 1\ \Omega$
- Branch 2 (Middle): $\text{EMF}_2 = 2\ \text{V}, r_2 = 1\ \Omega$
- Branch 3 (Bottom): $\text{EMF}_3 = 1\ \text{V}, r_3 = 1\ \Omega$

3. Calculation (Nodal Analysis Method)

Since the three branches are connected in parallel between points P and Q, they share common nodes.

Step A: Assumption of Potentials

Let the potential at point P be taken as reference: $V_P = 0\ \text{V}$

Let the potential at point Q be: $V_Q = V$ (unknown)

Step B: Application of Kirchhoff's Current Law (KCL)

As the circuit is isolated, the algebraic sum of currents at a node is zero.

Assuming currents flow from P to Q in all branches: $I_1 + I_2 + I_3 = 0$

Using Ohm's law: $I = \frac{\text{EMF} - V_Q}{r}$

Substituting for each branch: $\frac{3-V}{1} + \frac{2-V}{1} + \frac{1-V}{1} = 0$

Step C: Determination of Potential V

$$(3 - V) + (2 - V) + (1 - V) = 0$$
$$6 - 3V = 0;\ 3V = 6;\ V = \boxed{2\ \text{V}}$$

Thus, point Q is at a potential of 2 V with respect to point P.

Step D: Current in Branch 3

$$I_3 = \frac{\text{EMF}_3 - V}{r_3}$$
$$I_3 = \frac{1 - 2}{1} = -1\ \text{A}$$
$$\boxed{I_3 = -1\ \text{A}}$$

Conclusion

The negative sign indicates that the current in Branch 3 flows from Q to P.

Magnitude of current in Branch 3 $= 1$ A (from Q to P)

4. Correct Answers: B

$\boxed{T_2 > T_1}$ (Temperature T_2 is greater than temperature T_1

Explanation

Identification of Slope

The given graph is a plot of Current (I) on the Y-axis versus Voltage (V) on the X-axis.

From Ohm's law: $V = IR$ or $I = \frac{1}{R}V$
Hence, the slope of the I-V graph is:
slope $= \frac{I}{V} = \frac{1}{R}$
Steeper slope ⇒ Lower resistance ;

Smaller (shallower) slope ⇒Higher resistance

Comparison of Curves

The line corresponding to temperature T_1 is steeper than the line corresponding to T_2.

Therefore: $R_1 < R_2$
Relation with Temperature

For a metallic conductor, resistance increases with increase in temperature.

Since:

$$R_2 > R_1$$

it follows that:

$$T_2 > T_1$$

5. Correct Answers: D

Three resistors between X and A are in parallel, so their equivalent resistance is R/3. The upper branch becomes R/3 + R = 4R/3, while the lower branch is R. Since the total equivalent resistance is 2 Ω, solving gives R = 3.5 Ω. Using current division, the voltage across resistor r comes out to be 7.5 V.

6. Correct Answer: B) $P_2 : P_1 = 9 : 1$

Each quarter of the circular wire has resistance 1 Ω. Due to symmetry, no current flows through one diameter (QS), so it is ignored. Between P and R, there are three paths each of 2 Ω in parallel, giving equivalent resistance 2/3 Ω. Therefore, current = 5 ÷ (2/3) = 7.5 A.

7. Correct Answers: $\boxed{1.6\ \text{V}}$

Explanation

Parallel Connection

The bottom branch containing the 2 Ωand 3 Ωresistors is directly connected across the 4 Vbattery.
Therefore, the **total potential difference** across this branch is 4 V.

Series Connection

Within this branch, the 2 Ωand 3 Ωresistors are connected in **series**, so the applied voltage is divided between them.

Calculation (Voltage Division Rule) $V = 4 \times \frac{2}{2+3} = 4 \times \frac{2}{5}$ $V = \boxed{1.6\ \text{V}}$

5. MAGNECTIC EFFECTS OF ELCETRIC CURRENT

Solutions

Worksheet – 1

1. **Correct Answer: (B) A current-carrying conductor**

   Explanation: Magnetic field is produced only by moving charges, i.e., electric current.

2. **Correct Answer: (B) south to north**

   Explanation: Magnetic dipole moment of a bar magnet is directed from south pole to north pole inside the magnet

3. **Correct Answer: (B) 10 cm**

   Explanation: Magnetic length ≈ 0.84 × geometrical length = 0.84 × 12 ≈ 10 cm.

4. **Correct Answer: (B) 0.8 A·m²**

   Explanation: Magnetic moment = pole strength × magnetic length
   = 4 × 0.20 = 0.8 A·m².

5. **Correct Answer: (B) 2 T**

   Explanation: Magnetic flux density $B = \frac{\Phi}{A} = \frac{0.02}{0.01} = 2$ T.

6. **Correct Answer: (C) reverse direction of deflection**

   Explanation: Reversing current reverses the magnetic field direction, hence compass deflection reverses.

7. **Correct Answer: (C) Magnetic effect of electric current**

   Explanation: Magnetic field around a current-carrying wire verifies magnetic effect of current.

8. **Correct Answer: (D) 90°**

   Explanation: Magnetic flux $\Phi = BA\cos\theta$; when θ = 90°, cosθ = 0, so flux is zero.

9. **Correct Answer: (B) 0.75 A·m²**

   Explanation: Magnetic moment of a bar magnet is the product of its **pole strength** and **effective length**.

   $$M = m \times (2l)$$

   Here,
   $m = 5$ A mand $2l = 0.15$ m

   $$M = 5 \times 0.15 = 0.75 \text{ A m}^2$$

10. **Correct Answer: (C) one-fourth**

    Explanation: Magnetic force ∝ 1/r²; doubling distance reduces force to one-fourth.

WORKSHEET-2

1. **Correct Answer: (B) circular coil**

   Explanation: The diagram represents a magnetic field caused by a current-carrying conductor which is a circular coil

2. **Correct Answer: (D) uniform and parallel to the axis**

   Explanation: Inside a long solenoid, magnetic field lines are parallel and equally spaced.

3. **Correct Answer: (C) increasing field strength uniformly**

   Explanation: Magnetic field strength in a solenoid is directly proportional to current.

4. **Correct Answer: (B) all parts of the coil contribute field in the same direction**

   Explanation: At the centre, fields due to all segments add up, giving maximum field.

5. **Correct Answer: (C) current and distance from conductor**

   Explanation: Magnetic field around a straight wire depends on current and distance (B ∝ I/r).

6. **Correct Answer: (C) magnetic field is produced around the wire**

   Explanation: Current produces a magnetic field which deflects the compass needle.

7. **Correct Answer: (B) polarity of solenoid reverses**

   Explanation: Reversing current reverses the direction of magnetic field, hence polarity changes.

8. **Correct Answer: (B) Will increase**

   Explanation: Soft iron has high permeability, which strengthens the magnetic field.

9. **Correct Answer: (C) concentric circles**

   Explanation: Magnetic field lines around a straight current-carrying wire are circular.

10. **Correct Answer: (B) B ∝ I/r**

    Explanation: Magnetic field due to a straight conductor is directly proportional to current and inversely proportional to distance.

WORKSHEET-3

1. **Correct Answer: (C) Into the page**

   Explanation: The magnetic force is given by F = q (v × B). Here, the electron beam is moving downwards and the magnetic field is towards the right.
   For a positive charge, the force would be out of the page, but since an electron is negatively charged, the force acts in the opposite direction, i.e., into the page.

2. **Correct Answer: (C) Vertically downwards**

   Explanation: The magnetic field is directed from N to S. Using Fleming's Left Hand Rule with the given current direction, the thumb (force) points vertically downwards. Hence, the force on the wire is vertically downwards.

3. **Correct Answer: (C) parallel or anti-parallel to the magnetic field**

   Explanation: Magnetic force $F = BIL\sin\theta$; when conductor is parallel or anti-parallel, $\theta = 0°$or $180°$, so force is zero.

4. **Correct Answer: (D) direction of force on the conductor**

   Explanation: Fleming's left-hand rule gives the direction of force acting on a current-carrying conductor in a magnetic field.

5. **Correct Answer: (B) 0.4 N**

   Explanation: $F = BIL = 0.2 \times 4 \times 0.5 =$ 0.4 N.

6. **Correct Answer: (D)**

   Explanation: When conductor is parallel to field, $\boldsymbol{\theta = 0° \Rightarrow \sin\ 0 = 0 \Rightarrow F = 0}$

7. **Correct Answer: (B) reverse direction of force**

   Explanation: Direction of force depends on direction of magnetic field; reversing poles reverses force direction.

8. **Correct Answer: (C)**
   Explanation: $\boldsymbol{F = BIl\sin\ \theta = 0.2 \times 4 \times 0.5 \times \sin\ 90° = 0.4}$ N

9. **Correct Answer: (C) reverse current after every half rotation**

   Explanation: Commutator ensures continuous rotation by reversing current direction every half turn.

10. **Correct Answer: (B) plane of loop is perpendicular to field**

    Explanation: Torque $\tau = NBIA\sin\ \theta$; when plane is perpendicular, $\theta = 0°$, so torque is zero.

WORKSHEET-4

1. **Correct Answer: (C) opposition to change in magnetic flux**

   Explanation: Negative sign in Faraday's law represents Lenz's law, which opposes the change causing emf.

2. **Correct Answer: (D) conservation of energy**

   Explanation: Lenz's law ensures energy is conserved by opposing the change in flux.

3. **Correct Answer: (C) production of emf due to change in magnetic flux linked with a conductor**

   Explanation: This is the definition of electromagnetic induction.

4. **Correct Answer: (B) 5 Ω**

   Explanation: $R = \frac{V}{I} = \frac{10}{2} = 5\ \Omega$.

5. **Correct Answer: (A) 2 V**

   Explanation: emf $= \frac{\Delta\Phi}{\Delta t} = \frac{5-1}{2} = 2$ V.

6. **Correct Answer: (C) 10 V**

   Explanation: emf $= N\frac{\Delta\Phi}{\Delta t} = 50 \times \frac{0.02}{0.1} =$ 10 V.

7. **Correct Answer: (C) electromagnetic induction**

   Explanation: Security scanners detect induced currents in metal objects.

8. **Correct Answer: (B) 2 A**

   Explanation: $I = \frac{V}{R} = \frac{20}{10} = 2$ A.

9. **Correct Answer: (B) 10 V**

   Explanation: emf $= 100 \times \frac{0.05}{0.5} = 10$ V.

10. **Correct Answer: (C) emf does not depend on resistance**

    Explanation: Induced emf exists due to changing magnetic flux even if the circuit is open.

WORKSHEET-5

1. **Correct Answer: (C) coil completes half rotation**

   Explanation: After every half rotation of the coil in an AC generator, the direction of induced current reverses.

2. **Correct Answer: (C) Split-ring commutator**

   Explanation: A split-ring commutator converts alternating current into unidirectional (DC) current.

3. **Correct Answer: (C) 440 V**

   Explanation: $V_s/V_p = N_s/N_p = 2$; $V_s = 2 \times 220 = 440$ V.

4. **Correct Answer: (B) step-down transformer**

   Explanation: Secondary turns are fewer than primary turns, so voltage decreases.

5. **Correct Answer: (B) 220 V**

   Explanation: Standard domestic supply voltage between live and neutral in India is 220 V.

6. **Correct Answer: (C) each appliance gets same voltage**

   Explanation: Parallel connection ensures equal voltage across all appliances.

7. **Correct Answer: (C) 15 A circuit**

   Explanation: High-power appliances like geysers require higher current rating circuits.

8. **Correct Answer: (C) electrical energy consumed**

   Explanation: Electricity meter measures energy in kilowatt-hours (kWh).

9. **Correct Answer: (B) 2 : 1**

   Explanation: $E_s : E_p = N_s : N_p = 400 : 200 = 2 : 1$.

10. **Correct Answer: (C) overloading**

    Explanation: Connecting too many appliances to one socket draws excess current, causing overloading.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: (D) Both B and C**

   Explanation: In $F_m = BI\ell\sin\theta$, force F_m is perpendicular to both magnetic field B and length ℓ. Hence both pairs have right angle.

2. **Correct Answer: (D) Magnetic moment**

   Explanation: Magnetic moment is a vector quantity; all others are scalar.

3. **Correct Answer: (B)** $\frac{2M}{\pi}$

Explanation: When a magnet is bent into a semicircle, effective length becomes diameter. New magnetic moment $= \frac{2}{\pi}M$.

4. Correct Answer: (D) More for shorter magnet

Explanation: Magnetic moment $M = m \times 2l$. For same M, smaller length requires larger pole strength.

5. Correct Answer: (C) $\mathbf{1:2}$

Explanation: Magnetic moment $M \propto m \times l$
Given ratios: Length = 2: 3 ; Pole strength = 3: 4

$M_1 : M_2 = (3 \times 2) : (4 \times 3) = 6 : 12 = 1 : 2$

6. Correct Answer: (C) 25 A-m

Explanation: Magnetic moment $M =$ $m \times$ (distance between poles).
Given $M = 5$ A.m^2, distance = 20 cm = 0.20 m ; $\mathrm{m} = \dfrac{\mathrm{M}}{2\mathrm{l}}$

$$m = \frac{M}{0.20} = \frac{5}{0.20} = 25 \text{ A.m}$$

7. Correct Answer: (C) Remains unchanged

Explanation: Magnetic force $F \propto \frac{m_1 m_2}{r^2}$.
If distance is doubled → force becomes $\frac{1}{4}$.
If both pole strengths are doubled → force becomes 4times.
Net effect: no change.

8. Correct Answer: (B) 2 A-m

Explanation: $F = \frac{\mu_0}{4\pi}\frac{m_1 m_2}{r^2}$
Given: $F = 2 \times 10^{-7}$ N, $r = 1$ m, $\frac{\mu_0}{4\pi} = 10^{-7}$, test pole $m_1 = 1$ A.m

$2 \times 10^{-7} = 10^{-7} \times m \Rightarrow m = 2$ A.m

9. Correct Answer: (C) At rest

Explanation: Magnetic force $F = qvB\sin\theta$. If velocity $v = 0$, force is zero.

10. Correct Answer: (C) Both electric and magnetic fields

Explanation: A moving electric charge always produces an electric field and also a magnetic field.

11. Correct Answer: (A) Straight line

Explanation: When velocity is parallel to magnetic field, magnetic force $F = qvB\sin\theta = 0$. Hence proton moves in a straight line.

12. Correct Answer: (C) Circular in a plane perpendicular to conductor

Explanation: Magnetic field lines around a straight current-carrying conductor are concentric circles lying in planes perpendicular to the conductor.

13. Correct Answer: (B) 1 weber

Explanation: Magnetic flux $\Phi = BA\cos\theta$
Here $B = 4$ Wb m^{-2}, $A = 0.5$ m^2, angle between **B** and area vector $\theta = 60°$

$$\Phi = 4 \times 0.5 \times \cos 60° = 2 \times \frac{1}{2} = 1 \text{ weber}$$

14. Correct Answer: (c)

parallel to the magnetic field

15. Correct Answer: (C) $\pi/2$

Explanation: Force is maximum when angle between conductor and field is $90° = \pi/2$.

16. Correct Answer: (B) It will move outwards to plane of paper

Explanation: Using Fleming's left-hand rule:

Rule: We use Fleming's Left-Hand Rule to determine the direction of the force.

Forefinger (Field): Point your forefinger in the direction of the Magnetic Field (B), which is to the Right.

Center Finger (Current): Point your center finger in the direction of the Current (I), which is Upwards.

Thumb (Force): Your thumb will point Into the page (away from you).

Conclusion: The force acts inwards, perpendicular to the plane of the paper.

17. Correct Answer: (C) Away from you, into the page

Explanation: Magnetic field is from North to South (left → right).

Current flows upwards.
Using Fleming's Left-Hand Rule (or $\vec{F} = \vec{I} \times \vec{B}$), the force is directed into the plane of the paper, i.e., away from you.

18. Correct Answer: (A) 30°

Explanation: $F = BIL\sin\,\theta$

$$15 = 2 \times 10 \times 1.5 \times \sin\,\theta$$

$$15 = 30\sin\,\theta \Rightarrow \sin\,\theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

19. Correct Answer: (C) 1

Explanation: Magnetic force on a straight current-carrying conductor:

$$F = BIL\sin\,\theta$$

Segment ab is vertical, magnetic field B is horizontal $\Rightarrow \theta = 90^\circ$

$$F_{ab} = BIl$$

Segment bc makes 45°with ab, so angle with B is 45°.
Its length $bc = \frac{l}{\cos 45^\circ} = l\sqrt{2}$

$$F_{bc} = BI(l\sqrt{2})\sin\,45^\circ = BIl$$

$$\Rightarrow \frac{F_{bc}}{F_{ab}} = 1$$

20. Correct Answer: (B) 1.25 A·m²

Explanation: Magnetic moment:

$M = (\text{pole strength}) \times (\text{magnetic length})$

$M = 25 \times 0.05 = 1.25\text{A.m}^2$

21. Correct Answer: (B) 4 A·m and 16 A·m
Explanation: Force between magnetic poles:

$$F = \frac{\mu_0}{4\pi}\frac{m_1 m_2}{r^2}$$

Given:
Force = 64mg wt

$$= 64 \times 10^{-6}\,\text{kg} \times 10 = 6.4 \times 10^{-4}\,\text{N}$$

Distance $r = 10\ \text{cm} = 0.1\ \text{m}$

$$6.4 \times 10^{-4} = 10^{-7}\frac{m_1 m_2}{(0.1)^2} \Rightarrow m_1 m_2 = 64$$

One pole is 4 times the other $\Rightarrow m_2 = 4m_1$

$$4m_1^2 = 64 \Rightarrow m_1 = 4,\ m_2 = 16$$

22. Correct Answer: (C) 0.02 Wb

Explanation: Magnetic flux

$$\Phi = BA\cos\,\theta$$

Given:
Radius $r = 0.2$ m $\Rightarrow A = \pi r^2 = \pi(0.2)^2 = 0.04\pi\ \text{m}^2$
Magnetic field $B = \frac{1}{\pi}\ \text{Wb m}^{-2}$
Angle between axis (area vector) and field $\theta = 60^\circ$

$$\Phi = \frac{1}{\pi} \times 0.04\pi \times \cos\,60^\circ = 0.04 \times \frac{1}{2}$$

$$= 0.02\ \text{Wb}$$

23. Correct Answer: (A)

Explanation:
Torque on a magnet:

$$\tau = F \times d$$

where d =separation between poles.

Given:
Length = 20 cm $\Rightarrow$ pole separation = 0.20 m
Torque $\tau = 2 \times 10^{-5}$ N.m

$$F = \frac{\tau}{d} = \frac{2 \times 10^{-5}}{0.20} = 1 \times 10^{-4}\ \text{N}$$

But torque acts on **both poles**, so force on **each pole**:

$$F_{\text{each}} = 2 \times 10^{-4}\ \text{N}$$

24. Correct Answer: B) The current in each loop flows in opposite directions and D) The loops will attract each other

According to **Lenz's law**, the induced current in each loop will flow in a direction that opposes the change in magnetic flux causing it.

- As the magnet moves towards loop B (North pole first), the flux through B increases. The induced current in B creates a North pole on the left face to repel the incoming magnet, so the current flows counter-clockwise when viewed from the left (towards A).
- As the magnet moves away from loop A (North pole moving away), the flux through A decreases. The induced current in A creates a South pole on the

right face to attract the departing magnet, so the current flows counter-clockwise when viewed from the left (towards B).

- Relative to an external observer watching the magnet move from left to right, the current in loop A flows counter-clockwise and the current in loop B flows clockwise. Thus, the currents flow in **opposite directions** .
- Since the induced poles (South facing North) attract the magnet itself, the loops effectively create a force to resist the motion, meaning they are pulled towards the magnet, or **attract each other** in the context of the entire system resisting the change in motion

25. Magnetic field at the centre = 2 × 10^{-2} T ((A),(C)

Explanation : Magnetic field at the centre of a circular coil is

$$B = \frac{\mu_0 NI}{2R}$$

Given:
Radius, $R = \pi$ cm $= \pi \times 10^{-2}$ m
Number of turns, $N = 50$
Current, $I = 20$ A

$$\mu_0 = 4\pi \times 10^{-7}$$

$$B = \frac{4\pi \times 10^{-7} \times 50 \times 20}{2\pi \times 10^{-2}} = 2 \times 10^{-2} \text{ T}$$

26. Distance from the wire = 4 m (A),(D)

Explanation: Magnetic field due to a long straight current-carrying conductor is

$$B = \frac{\mu_0 I}{2\pi r}$$

Given:

$$I = 10 \text{ A}$$
$$B = 5 \times 10^{-7} \text{ T}$$
$$\mu_0 = 4\pi \times 10^{-7}$$

$$r = \frac{\mu_0 I}{2\pi B} = \frac{4\pi \times 10^{-7} \times 10}{2\pi \times 5 \times 10^{-7}} = 4 \text{ m (400cm)}$$

27. Correct Answer: (A) Electrical energy

Explanation: A generator converts mechanical energy into electrical energy based on electromagnetic induction.

28. Correct Answer: (B)

29. Correct Answer: (D)

Explanation: Assertion (A) is FALSE
Magnetic field lines around a current-carrying circular conductor (coil) are not concentric circles everywhere; this description is correct for a straight conductor, not a circular loop.

Reason (R) is TRUE
A current-carrying straight conductor produces circular magnetic field lines around it.

30. Correct Answer: (A)

Explanation: Assertion (A) is TRUE
Force is maximum when the conductor is perpendicular to the magnetic field.

Reason (R) is TRUE
Since $F = BIL\sin\ \theta$, force depends on angle and is maximum at 90°. R correctly explains A.

31. Correct Answer: (C)

Explanation: Assertion (A) is TRUE
Induced current is produced when magnetic flux linked with a coil changes.

Reason (R) is FALSE
Electromagnetic induction can occur due to changing current, changing area, or changing field, not only by moving a magnet.

32. Correct Answer: (D)

Explanation: Assertion (A) is FALSE
An electric generator converts mechanical energy into electrical energy, not magnetic energy.

Reason (R) is TRUE
A generator works on the principle of electromagnetic induction.

33. Correct Matching

(a) → Q
(b) → R
(c) → P
(d) → S

Explanation:

- When a conductor is parallel to the magnetic field, angle $\theta = 0°$, so $F = BIL\sin\ \theta = 0$→ No force.
- When a conductor is perpendicular to the magnetic field, $\theta = 90°$, force is maximum.
- Increasing the speed of rotation increases rate of change of flux → maximum induced emf.
- Moving a magnet faster increases rate of change of magnetic flux → increase in induced current.

34. Correct Matching

(a) Fleming's Left Hand Rule → Q
(b) Fleming's Right Hand Rule → P
(c) Electromagnetic induction → R
(d) Transformer → S

Explanation:

- Fleming's Left Hand Rule is used to find the direction of force → Electric motor.
- Fleming's Right Hand Rule is used to find the direction of induced current → generator principle.
- Electromagnetic induction refers to the production of induced current due to change in magnetic flux.
- Transformer works on AC and is used for change in AC voltage.

PASSAGE -1 Given (from passage):

- Transformer is ideal → no power loss
- Turns ratio $N_p : N_s = 2:5$
- Primary current $I_p = 10$ A
- Input power $P_{\text{in}} = 300$ W

Step-1: Find Primary Voltage

$$P = V_p I_p \Rightarrow V_p = \frac{300}{10} = 30 \text{ V}$$

35. Correct Answer: The current through the secondary coil would be —

Correct Answer: (C) 4 A

Explanation: $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{5}{2} \Rightarrow V_s = \frac{5}{2} \times 30 = 75$ V
For ideal transformer: $P_{\text{out}} = P_{\text{in}} = 300$ W $\Rightarrow I_s = \frac{300}{75} = 4$ A

36. Correct Answer: The voltage in the secondary coil would be —

Correct Answer: (D) 75 V

Explanation: $V_s = \frac{N_s}{N_p} V_p = \frac{5}{2} \times 30 = 75$ V

37. Correct Answer: The output power would be — Correct Answer: (C) 300 W

Explanation: In an **ideal transformer**, output power equals input power.

$$P_{\text{out}} = P_{\text{in}} = 300 \text{ W}$$

PASSAGE-2 Given (from PASSAGE):

- Two long parallel straight wires
- $i_1 = 15$ A upward (left wire)
- $i_2 = 30$ A downward (right wire)
- Distance between wires = 15 cm
- Point P is 5 cm to the right of left wire, hence 10 cm to the left of right wire

Formula used: $B = \frac{\mu_0 I}{2\pi r} = \frac{2\times10^{-7} I}{r}$

38. Correct Answer: Magnitude of resultant magnetic field at point P

Field due to left wire ($i_1 = 15\ A$):

$$r_1 = 5 \text{ cm} = 0.05 \text{ m}$$
$$B_1 = \frac{2 \times 10^{-7} \times 15}{0.05} = 6 \times 10^{-5} \text{ T}$$

Field due to right wire ($i_2 = 30\ A$):

$$r_2 = 10 \text{ cm} = 0.10 \text{ m}$$
$$B_2 = \frac{2 \times 10^{-7} \times 30}{0.10} = 6 \times 10^{-5} \text{ T}$$

Direction check:
Using right-hand thumb rule, both

fields are in the same direction at P, so they add.

$B_{\text{net}} = B_1 + B_2 = 12 \times 10^{-5} = 1.2 \times 10^{-4}$ T
Correct Answer: (A) 1.2×10^{-4} T

39. Correct Answer: Direction of the magnetic field at point P

- Field due to left wire (current upward): into the plane
- Field due to right wire (current downward): into the plane

Correct Answer: (B) Into the plane of the paper

40. If the direction of i_2 is reversed, magnetic field at P would be

- Now both currents are upward
- At point P:
- Field due to left wire → into the plane
- Field due to right wire → out of the plane
- Magnitudes are equal: $B_1 = B_2 = 6 \times 10^{-5}$ T

They cancel each other. $B_{\text{net}} = 0$

41. Correct Answer: 8×10^{-7} T

Explanation: Magnetic field near a straight conductor is **directly proportional to current**.
If current is doubled (2 A → 4 A), magnetic field also doubles:

$$4 \times 10^{-7} \text{ T} \rightarrow 8 \times 10^{-7} \text{ T}$$

42. Correct Answer: 0 N

Explanation: Magnetic force on a current-carrying conductor: $F = BIL\sin\theta$; When conductor is parallel to magnetic field, $\theta = 0°$, so $\sin 0 = 0$. Hence, force = 0 N.

43. Correct Answer: 2 times

Explanation: In an AC generator, the direction of induced current reverses after every half rotation.
So, in one complete rotation, the direction changes twice.

Previous question papers

1. Correct Answer: B. (ii) and (iii) only

Explanation: For current-carrying conductors:

- Same direction of current → attraction
- Opposite direction of current → repulsion

From the figure:

- Part I: current upward
- Part II: current downward
- Part III: current downward
- Part IV: current upward

Check each statement:

- I and IV → same direction → attract (false)
- II and IV → opposite directions → repel (true)
- I and III → opposite directions → repel (true)
- II and III → same direction → attract (false)

2. Correct Answer: B Option B: (i) T (ii) F (iii) T (iv) F

Explanation:

- True – Soft iron increases magnetic permeability, so the magnetic field inside a solenoid increases.
- False – By the right-hand thumb rule, for current upward, the magnetic field is anticlockwise when viewed from above, not clockwise.
- True – Reversing current reverses polarity, not the strength, if current magnitude is same.
- False – Soft iron is best for electromagnet cores, not steel (steel retains magnetism).

3. Correct Answer: C Option C: II only

Explanation:

- A current-carrying solenoid produces a magnetic field pattern exactly like a bar magnet (distinct north and south poles).
- Straight wire and loops do not fully resemble a bar magnet field pattern.

4. Correct Answer: At the 3 cm mark

Explanation: Magnetic field due to a long straight current-carrying conductor:

$$B = \frac{\mu_0 I}{2\pi r}$$

For zero magnetic field, fields due to both conductors must be equal and opposite. Since currents are in the same direction, the neutral point lies between the two wires and closer to the smaller current (1 A).

- Wire I at 2 cm, current = 1 A
- Wire II at 6 cm, current = 3 A

Let the neutral point be at $x$cm.

$$\frac{1}{x-2} = \frac{3}{6-x}$$
$$6 - x = 3(x - 2)$$
$$6 - x = 3x - 6$$
$$4x = 12 \Rightarrow x = 3 \text{ cm}$$

5. Correct Answer:

6. Correct Answer: A

Small Explanation:

- A straight current-carrying wire produces a magnetic field, so a nearby compass needle shows **deflection → True**.

7. Correct Answer: C

magnetic field will be in same direction with equal magnitude because of the incoming and outgoing currents so net effect will not be zero at the center in case(c)

8. Correct Answer: A

According to clock rule, the current is in anticlockwise direction in the uppers face of the loop, so the upper face acts as north pole.